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基於 RSSI 的自適應路徑損耗模型之

室內定位系統

Indoor Positioning System Using an Adaptive
Path Loss Model Based on RSSI

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摘要

在基於 RSSI 的室內定位系統中，對數距離路徑損耗模型（Log-distance Path Loss Model, LDPL）廣泛應用於距離預估。然而，傳統模型通常假設環境衰減因子（Path Loss Exponent） n 為固定值，忽略不同空間區域中無線訊號傳播特性的差異，導致距離估算誤差偏高，進而影響定位精度。為解決此問題，本研究提出一種自適應路徑損耗模型，可根據環境特性動態調整每個目標點的 n 值，提升模型對不同場域條件的適應能力與定位準確度。

本研究使用 ESP32 Wi-Fi 模組作為無線通訊設備，於 $11\text{m} \times 15\text{m}$ 的室內場域中部署四個裝置作為錨節點，以蒐集大量 RSSI 數據。資料處理方面，首先利用卡爾曼濾波器去除雜訊與突變值，以提升訊號穩定性。接著，根據錨節點間的 RSSI 關係，模擬並推估出每個目標點的自適應路徑損耗指數 n 。最後，透過估算的 n 值計算目標點與各錨節點間的距離，並以最小平方法（Least Squares Method）推估出目標點的二維座標位置。

研究結果顯示，本系統在 20 個目標點進行驗證的平均絕對誤差（Mean Absolute Error, MAE）為 2.33 公尺；在 10 個獨立測試點進行驗證時，其 MAE 為 1.88 公尺，相較於傳統現場校準法（MAE 為 3.25 公尺）與指紋定位法中表現最佳的隨機森林模型（MAE 為 2.65 公尺），皆展現出明顯優勢。CDF 累積分佈函數曲線亦顯示，在各誤差閾值下，本研究方法均能達到更高的定位準確度百分比。此外，與近期相關研究相比，本方法於使用錨節點更少、測試點更多的情況下，仍能達到 2.11 公尺的平均誤差，展現良好的定位表現與應用潛力。

關鍵詞：室內定位、Received Signal Strength Indicator、自適應路徑損耗模型、卡爾曼濾波、多邊定位

Abstract

In RSSI-based indoor positioning systems, the Log-distance Path Loss Model (LDPL) is widely used for distance estimation. However, traditional models typically assume a fixed Path Loss Exponent (n), neglecting the variability in wireless signal propagation characteristics across different spatial regions. This simplification often leads to significant errors in distance estimation and, consequently, reduced positioning accuracy. To address this issue, this study proposes an adaptive path loss model that dynamically adjusts the value of n for each target point based on environmental characteristics, thereby improving the model's adaptability and positioning precision in diverse indoor conditions.

This research employs ESP32 Wi-Fi modules as wireless communication devices and deploys four of them as anchor nodes within an $11\text{m} \times 15\text{m}$ indoor environment to collect a large volume of RSSI data. In terms of data processing, a Kalman filter is first applied to remove noise and sudden fluctuations, thereby enhancing signal stability. Then, based on the RSSI relationships among anchor nodes, the adaptive path loss exponent n for each target point is estimated through simulation. Finally, the estimated n values are used to calculate the distances between the target and each anchor node, and the target's 2D coordinates are determined using the Least Squares Method.

Experimental results show that the proposed system achieved an average positioning error of 2.33 meters over 20 target points. In a separate evaluation with 10 independent test points, the Mean Absolute Error (MAE) was 1.88 meters—significantly outperforming the traditional field-calibration method (MAE of 3.25 meters) and the best-performing fingerprinting method using Random Forest (MAE of 2.65 meters). The Cumulative Distribution Function (CDF) curves also demonstrate that the proposed method consistently achieves higher positioning accuracy across all error thresholds. Furthermore, compared to a recent related study, the proposed method achieves an

average error of 2.11 meters despite using fewer anchor nodes and more test locations, highlighting its strong performance and practical potential for real-world deployment.

Keywords: Indoor Positioning, Received Signal Strength Indicator, Adaptive Path Loss Model, Kalman Filter, Multilateration



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Chapter 1 Introduction

This chapter describes the motivation, contributions, and organization of this study.

1.1 Motivation

With the rapid development and widespread adoption of the Internet of Things (IoT), the demand for Location-Based Services (LBS) has experienced explosive growth. This surge not only profoundly influences our daily lives but also brings unprecedented transformations and opportunities across various industries. Traditionally, the Global Positioning System (GPS), as the most mature and widely applied positioning technology, has demonstrated irreplaceable value in outdoor environments. However, once entering indoor settings, GPS signals are significantly weakened or even completely blocked due to factors such as structural obstructions, multipath effects, and signal attenuation. As a result, GPS suffers from severe accuracy degradation or complete failure in indoor environments. Therefore, the development of efficient, accurate, and robust indoor positioning technologies suitable for complex indoor scenarios has become a major focus and challenge in both academia and industry.

Indoor positioning technologies have an extremely wide range of application scenarios, and the demand continues to grow. In healthcare and elderly care institutions, precise indoor positioning systems can help caregivers track the location of patients or elderly individuals in real time—for example, detecting falls, issuing alerts for wandering into restricted areas, or tracking valuable mobile medical assets such as wheelchairs—thereby greatly improving care efficiency and safety. In large public spaces such as airports, train stations, shopping malls, and museums, indoor navigation

systems can provide accurate route guidance to prevent disorientation, optimize pedestrian flow, and offer efficient evacuation assistance in emergencies to ensure safety. In the retail industry, indoor positioning enables customer flow analysis and hotspot detection and allows for the delivery of personalized advertisements and services based on real-time customer location, enhancing the shopping experience and generating greater business value. Moreover, in industrial applications such as smart factories, warehouse management, and underground mining, indoor positioning plays a key role in personnel tracking, equipment monitoring, and asset management, thereby improving operational efficiency and safety. These diverse and growing needs collectively drive continuous innovation and market expansion in indoor positioning technologies. According to the latest report published by Technavio in May 2025, as illustrated in Figure 1.1, the indoor positioning and indoor navigation (IPIN) market is undergoing explosive growth[1]. From 2024 to 2029, the market is expected to expand by USD 403.83 billion, with a compound annual growth rate (CAGR) as high as 72.2%. This remarkable growth is mainly attributed to the proliferation of IoT devices and the rapid advancement of wireless communication technologies such as Wi-Fi, Bluetooth Low Energy (BLE), and Ultra-Wideband(UWB).



Figure 1.1 Indoor Positioning and Indoor Navigation (IPIN) Market: Strategic Insights and Forecast (2025–2029)

Despite the growing market demand and rapid technological advancements, different application scenarios impose distinct requirements on the performance, cost, and deployment complexity of indoor positioning technologies. For instance, in sensitive environments such as hospitals, there are strict regulations regarding electromagnetic radiation. This necessitates minimizing the number of wireless communication devices in the positioning system to reduce potential electromagnetic interference. Additionally, certain settings are unsuitable for time-consuming and large-scale calibration processes, such as collecting signal strength data and constructing fingerprint databases. These practical concerns present major challenges for the implementation of indoor positioning technologies. Therefore, this study aims to identify a method that can improve indoor positioning accuracy while minimizing both the number of wireless devices and the extent of manual calibration required.

Among the various indoor positioning technologies, those based on radio frequency (RF) have attracted considerable attention due to their widespread adoption and relatively low cost. These technologies include Wi-Fi, Bluetooth, ZigBee, RFID, and Ultra-Wideband (UWB) [2]. Among them, Wi-Fi and Bluetooth are the most commonly used due to their high availability and cost-effectiveness. Compared with Bluetooth, Wi-Fi modules tend to deliver stronger and more stable signal strengths over both short and long distances. Therefore, this thesis adopts Wi-Fi as the wireless sensing medium.

Indoor positioning algorithms can generally be divided into two categories: model-based methods and scene analysis-based methods. Model-based approaches estimate the target's position using signal parameters collected at the receiver side, such as Time of Arrival (TOA), Time Difference of Arrival (TDOA), Angle of Arrival (AOA), or Received Signal Strength Indicator (RSSI) [3]. In contrast, scene analysis-based methods, typically represented by fingerprinting, involve collecting environmental signal features at numerous reference points during the offline stage to build a fingerprint database[4]. During the online stage, real-time measurements are compared against this database to estimate the target's position.

From the perspective of signal characteristics, TOA and TDOA methods require precise time synchronization and complex time measurement systems, making them costly. AOA-based methods need additional hardware such as antenna arrays, increasing system complexity and cost. In comparison, RSSI-based methods have gained popularity due to their simplicity and low implementation cost. One common RSSI-based technique is multilateration (or trilateration), which estimates the position of a target by calculating distances from the target to multiple known reference points (anchor nodes) using RSSI values and then applying geometric computation[5]. This method does not

require sophisticated hardware, making it a cost-effective and flexible indoor positioning approach.

However, the main challenge of RSSI-based multilateration lies in its limited accuracy. RSSI values are highly susceptible to fluctuations caused by multipath effects and environmental interferences, which affect the reliability of distance estimations. Multilateration relies on the path loss model [6] to convert RSSI values into distances, typically expressed as:

$$RSSI(d) = RSSI(d_0) - 10n\log_{10}(d) + X_{\sigma} \quad (1.1)$$

In this model, n is a crucial parameter that characterizes the signal attenuation rate in a specific environment. It is generally assigned an empirical value between 2 and 5. However, due to the complexity and heterogeneity of indoor environments, applying a fixed empirical value or conducting large-scale manual calibration to determine n often fails to reflect the actual signal propagation characteristics under varying spatial and temporal conditions. The relationship between RSSI and distance is logarithmic, meaning that even a slight deviation in the n value can result in significant errors in distance estimation. For example, given the same RSSI value, distance estimations may vary severalfold depending on whether n is set to 2 or 5. This is one of the critical reasons for the low accuracy in RSSI-based positioning.

Hence, the core motivation of this study is to address this issue by proposing an innovative solution that enables the automatic estimation of the path loss exponent n . This approach aims to reduce the need for manual calibration while improving positioning accuracy, thereby achieving a more accurate, flexible, and easily deployable indoor positioning system. The expected outcomes of this research may provide new

insights and practical references for the future development of indoor positioning technologies, particularly in scenarios with strict constraints on cost, deployment complexity, and electromagnetic emissions.

1.2 Contributions

This study addresses the common challenges of RSSI (Received Signal Strength Indicator) instability and the heavy reliance on site-specific calibration in indoor positioning applications. An adaptive path loss model based on RSSI is proposed, and an experimental platform is built to validate its effectiveness. The designed system is practical, accurate, and low-cost, meeting diverse indoor positioning needs. The main contributions of this research are summarized as follows:

■ **Proposal of an adaptive path loss model based on RSSI**

This study improves the conventional log-distance path loss model, which requires manual specification of environmental parameters, by introducing a method to automatically estimate the path loss exponent n . The proposed approach leverages relative RSSI data between multiple anchor nodes and uses the BFGS (Broyden–Fletcher–Goldfarb–Shanno) algorithm to minimize the error between predicted and actual distances, enabling dynamic adjustment of model parameters to better match the real-time wireless environment of the target node.

■ **Development of a modular experimental platform and real-world validation**

To validate the feasibility and practicality of the proposed model, a modular, portable indoor positioning system was developed based on ESP32 Wi-Fi modules. This platform features low cost, ease of assembly, and flexibility for expansion and

deployment. Through extensive data collection at different locations and time points, the experimental results demonstrate that the system performs robustly and stably under real-world conditions.

■ **Significant improvement in indoor positioning accuracy**

The proposed method demonstrates strong performance in terms of positioning accuracy. In comprehensive tests conducted on 20 different target locations, the system achieved a mean absolute error (MAE) of 2.33 meters. In a separate evaluation involving 10 independent test points, the MAE further decreased to 1.88 meters. This outperforms the traditional calibration-based method (3.25 meters) and the best-performing fingerprinting-based machine learning model, Random Forest (2.65 meters). In addition, cumulative distribution function (CDF) plots show that the proposed method consistently achieves the highest accuracy across various error thresholds.

■ **Reduction in deployment cost and enhancement of system flexibility**

Another important contribution of this study is the significant simplification of the system deployment process. Traditional fingerprinting methods require extensive site surveys and regular maintenance of the fingerprint database, resulting in high deployment and operational costs. The proposed method adopts adaptive modeling and real-time RSSI estimation, eliminating the need for large-scale environmental calibration or manual data labeling. This greatly reduces labor and resource requirements in the early stages and improves the system's deployment efficiency and scalability.

1.3 Organization

Chapter 1 introduces the motivation and background of this study, along with a preliminary overview of its key contributions. Chapter 2 provides the necessary background on indoor positioning technologies. It systematically reviews existing techniques and focuses on the research progress of model-based and scene analysis-based positioning methods. Chapter 3 presents the architecture of the proposed indoor positioning system. This includes the data collection process, data processing methods, and the detailed structure and procedures of the adaptive indoor positioning method developed in this study. Chapter 4 describes the experimental results obtained from real-world testing environments. The performance of the adaptive indoor positioning method is evaluated across 20 target locations and compared with other positioning methods at 10 additional independent test points. Moreover, a comprehensive comparison is conducted between this study and other related works based on RSSI-based multilateration. Chapter 5 summarizes the research findings and discusses potential directions for future work and possible areas of improvement.

Chapter 2 Related Work

This chapter aims to provide a comprehensive review and analysis of the background knowledge and related literature on indoor positioning technologies.

2.1 Indoor Positioning Technologies

Due to the diverse application scenarios and complex environmental characteristics of indoor settings, indoor positioning technologies have evolved into various classification schemes. In this study, the mainstream indoor positioning methods are categorized and summarized based on their positioning principles and the technologies employed. Figure 2.1 [7] illustrates a typical classification tree of indoor positioning technologies, which is primarily divided into three categories: commonly used indoor positioning technologies, indoor positioning signal properties, and indoor positioning algorithms.

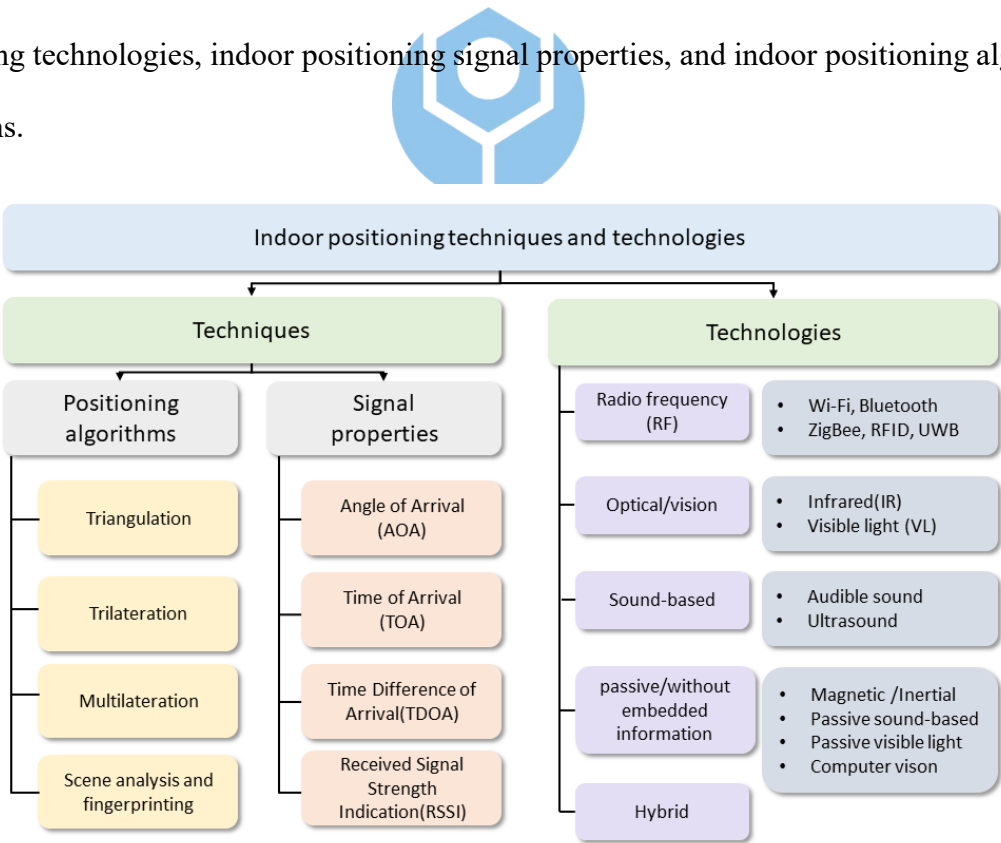


Figure 2.1 Classification of indoor positioning techniques and technologies.

Wireless communication technologies form the foundation of indoor positioning systems, serving as the means to transmit and receive signals used for localization. As shown in Figure 2.1, these technologies primarily fall within the domain of Radio Frequency (RF), with Wi-Fi, Bluetooth, Ultra-Wideband (UWB), RFID, and ZigBee being the most representative examples. Wi-Fi, as one of the most widely adopted wireless communication technologies, is commonly used in indoor positioning due to its widespread infrastructure, ease of deployment, and low cost [8][9]. Bluetooth, particularly Bluetooth Low Energy (BLE), is widely applied in IoT devices because of its low power consumption and flexible deployment [10][11], although it has a shorter signal range and generally lower positioning accuracy compared to technologies like UWB. Ultra-Wideband (UWB) offers centimeter-level positioning accuracy due to its extremely wide bandwidth and narrow pulse signal characteristics, and it effectively mitigates multipath effects [12]. However, its adoption is limited by higher costs and regulatory restrictions. RFID is suitable for short-range, passive tag scenarios but requires dense deployment of readers. ZigBee features low power consumption and mesh networking capabilities, making it common in smart home applications, though its positioning use is limited and generally confined to small areas. In addition to the aforementioned RF-based technologies, indoor positioning has also explored alternative non-RF signal sources. For instance, Visible Light Communication (VLC) utilizes LED lighting to transmit signals, offering high positioning accuracy and immunity to electromagnetic interference, though it is sensitive to lighting conditions and obstructions. Ultrasonic positioning estimates location based on sound wave propagation time and provides high accuracy, but it has poor penetration capability. Geomagnetic positioning leverages local magnetic field anomalies and requires no additional transmitters, but it relies on the

construction of magnetic maps and is susceptible to metal interference. Inertial Measurement Units (IMUs) use accelerometers and gyroscopes to perform short-term pedestrian dead reckoning; however, they suffer from accumulated errors and typically need to be fused with other technologies to improve stability.

Each technology comes with its own strengths and limitations, and the choice of method depends on the specific application requirements, environmental conditions, and trade-offs between accuracy, cost, and deployment complexity.

2.2 Indoor Positioning Signal properties

This section introduces several commonly used signal characteristics in indoor positioning, including Time of Arrival (TOA) [3], Time Difference of Arrival (TDOA) [3], Angle of Arrival (AOA) [13], and Received Signal Strength Indicator (RSSI).

2.2.1 Time of Arrival (TOA)

TOA is a distance estimation technique based on the signal propagation time. The fundamental principle involves accurately measuring the total time it takes for a wireless signal to travel from the transmitter to the receiver. Since wireless signals propagate through the air at a speed close to the speed of light c , the distance d between the transmitter and receiver can be calculated from the signal propagation time t . The mathematical relationship is given by:

$$d = c \times t \quad (2.1)$$

The advantage of the TOA technique is its potential to deliver high-accuracy ranging results. However, it requires extremely precise time synchronization across the

system. All receivers and the transmitter must achieve synchronization with nanosecond-level precision. In addition, multipath effects and non-line-of-sight (NLOS) propagation can distort signal delay measurements, leading to reduced ranging accuracy.

2.2.2 Time Difference of Arrival (TDOA)

Time Difference of Arrival (TDOA) is a variant of the TOA technique. Instead of measuring the absolute arrival time of the signal, it measures the difference in arrival times at multiple receivers. By using at least three receiving nodes, a hyperbolic positioning model can be constructed based on the time differences, where the intersection point of the hyperbolas indicates the target's location. The mathematical relationship can be expressed as:

$$d_i - d_j = c \times (t_i - t_j) \quad (2.2)$$

where d_i and d_j represent the distances from the target node to anchor nodes i and j , and t_i and t_j are the corresponding signal arrival times. Compared to TOA, TDOA reduces the system's synchronization requirements. It only requires precise time synchronization between the receivers, not with the transmitter. However, it still suffers from interference caused by multipath propagation and non-line-of-sight (NLOS) conditions. Moreover, solving the hyperbolic equations typically involves higher computational complexity.

2.2.3 Angle of Arrival(AOA)

Angle of Arrival (AOA) determines the position of a transmitter by measuring the angle at which a wireless signal arrives at the receiver. Implementing AOA typically requires receivers equipped with specialized antenna arrays, such as phased arrays or

smart antennas. These arrays estimate the signal's arrival angle by analyzing differences in phase, time, or signal strength between multiple antenna elements.

Once the AOA is determined at two or more known anchor nodes, the position of the target node can be calculated using triangulation. If the receiver is located at (x_R, y_R) and the transmitter is at (x_T, y_T) , the arrival angle θ can be approximated as:

$$\theta = \arctan \left(\frac{y_T - y_R}{x_T - x_R} \right) \quad (2.3)$$

This method does not require distance measurements, as it directly utilizes angle information for positioning. However, implementing AOA requires additional hardware support. Furthermore, signals arriving via reflected paths may cause significant deviations in the estimated angle, thereby reducing positioning accuracy.

2.2.4 Received Signal Strength Indicator (RSSI)

RSSI refers to the signal power measured by a wireless receiver, and it is one of the most commonly used positioning features. Based on the physical characteristic that signal strength attenuates with distance, RSSI can be converted into a distance estimate using a path loss model, which is then combined with trilateration or multilateration to infer the position. The advantages of this method include high cost-effectiveness, as most existing wireless communication devices have built-in RSSI measurement capabilities, eliminating the need for additional specialized hardware and thus reducing deployment costs. It also offers deployment flexibility: compared to the stringent time synchronization requirements of TOA/TDOA or the need for antenna arrays in AOA, RSSI only requires simple signal transceivers and is easy to acquire data. Detailed RSSI signal characteristics and challenges will be elaborated in Section 2.5.1.

2.3 Indoor Positioning Algorithms

Indoor positioning algorithms are mathematical and computational methods that transform the aforementioned signal features into actual spatial coordinates. This section introduces several commonly used algorithms in indoor positioning systems, including trilateration and multilateration [5], triangulation [14], and fingerprinting [4].

2.3.1 Trilateration and Multilateration

Trilateration is a special case of Multilateration. Its principle involves using distance information between three known anchor nodes and a target node to determine the target node's spatial coordinates. In an ideal scenario, as shown in Figure 2.2, if the distances d_1, d_2, d_3 from the target node to the three known anchor nodes can be precisely measured, the target node's position can be determined as the intersection point of three circles, each centered at an anchor node with the corresponding distance as its radius. Assuming the unknown coordinates of the target node are (x, y) , and the coordinates of the three anchor nodes are (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . the following system of equations can be set up based on the Pythagorean theorem for solving:

$$(x - x_1)^2 + (y - y_1)^2 = d_1^2 \quad (2.4)$$

$$(x - x_2)^2 + (y - y_2)^2 = d_2^2 \quad (2.5)$$

$$(x - x_3)^2 + (y - y_3)^2 = d_3^2 \quad (2.6)$$

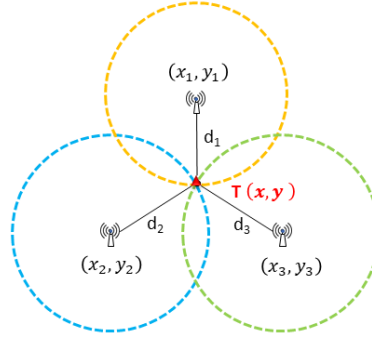


Figure 2.2 Trilateration positioning method.

2.3.2 Triangulation

In triangulation, the method of computing a position varies depending on the system architecture, and it can be categorized into two approaches: remote positioning and self-positioning, as illustrated in Figure 2.3. In Figure 2.3(a), which shows remote positioning, at least two reference nodes, (x_1, y_1) and (x_2, y_2) , measure the angles of arrival θ_1 , θ_2 from the target node T. These angle measurements are transmitted to a remote server, which estimates the position of the target node (x, y) based on the intersection point of the angle lines. As shown in Figure 2.3(b), self-positioning involves the target node T itself measuring the angles θ_{12} , θ_{23} , θ_{13} , with respect to three

known reference nodes. Using these angles and the coordinates of the reference nodes, the target node can calculate its own position based on sine and cosine relationships.

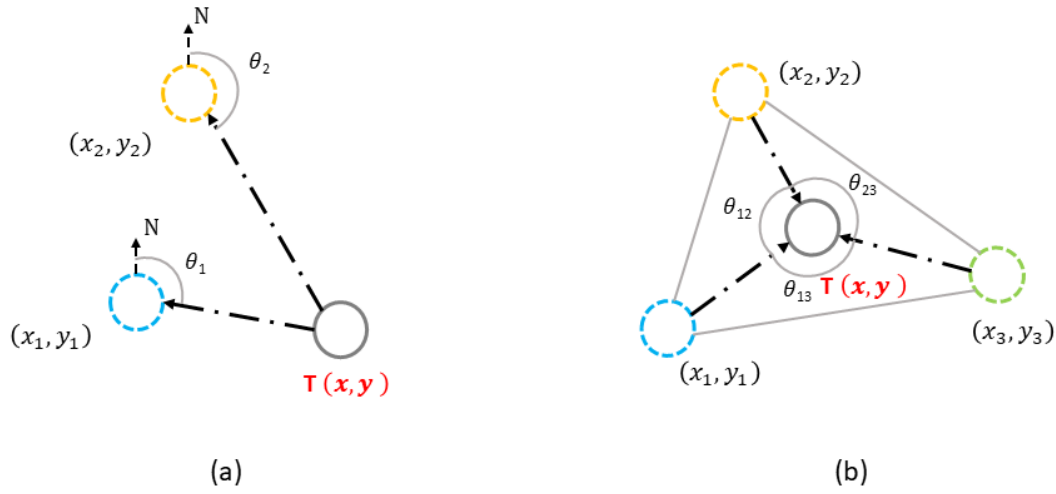


Figure 2.3 Triangulation positioning method. (a) Remote positioning. (b) Self-positioning.

2.3.3 Fingerprinting

Fingerprinting is a method that determines position by analyzing the distribution of wireless signal features in an environment. This approach is typically divided into two phases: the offline phase and the online phase. In the offline phase, the system selects multiple reference points within the indoor space and collects signal features from surrounding wireless devices at each point. These feature data are paired with their corresponding physical coordinates to build a positioning database, also known as a radio map. During the online phase, when a client device needs to determine its location, it collects real-time signal features from the current environment and compares them with the entries in the radio map. The system then estimates the user's actual position based on similarity measures or predictive models.

The advantages of fingerprinting include high accuracy and the fact that it does not require complex hardware. When the signal distribution is stable and the fingerprint database is well maintained, this method can achieve high positioning precision. However, its main challenges include signal instability, sensitivity to environmental changes, time-consuming data collection during the offline phase, and the high cost of maintaining the fingerprint database.

2.4 RSSI-Based Indoor Positioning Studies

This section outlines the characteristics and challenges of RSSI signals, wireless signal propagation models, and commonly used filtering methods, providing an overview of existing research strategies for addressing the instability of RSSI in indoor positioning applications.

2.4.1 Characteristics and Challenges of RSSI

RSSI measures the power of a wireless signal upon reaching the receiver. However, in indoor environments, it is highly susceptible to interference, leading to a significant decrease in positioning accuracy. Multipath effects in indoor spaces cause signals to reflect, refract, and diffract off walls, furniture, or human bodies, creating multiple propagation paths. When these signals superimpose, they cause RSSI values to fluctuate randomly in a short period, making it difficult to accurately correspond to the actual distance. Shadowing occurs when the signal propagation path is blocked by large obstacles (such as thick walls, metal cabinets, or human bodies), causing random attenuation of power. Even at the same distance, RSSI values may vary significantly at different locations, and this type of attenuation often follows a log-normal distribution.

Furthermore, indoor environments mostly involve non-line-of-sight (NLOS) propagation, where signals that bypass or penetrate obstacles experience more severe attenuation and longer propagation times, leading to lower RSSI readings and consequently overestimating the actual distance. Electromagnetic interference from other wireless devices (such as Wi-Fi, Bluetooth, microwave ovens, etc.) also affects RSSI measurements, causing signal fluctuations and reducing reliability. Human activity is also a significant factor, as the human body absorbs, reflects, and scatters wireless signals; personnel movement or obstruction of signals can cause rapid changes in RSSI, further exacerbating signal instability. The aforementioned complex environmental factors collectively lead to the high volatility and unpredictability of RSSI measurements, which is a core challenge faced by RSSI-based indoor positioning systems.

2.4.2 Wireless Signal Propagation Model

The Free Space Path Loss (FSPL) model [15][16] is the most fundamental wireless propagation model used to describe the attenuation of signal power as it travels from a transmitter to a receiver in an unobstructed environment. This model assumes that signals propagate only along a direct Line-of-Sight (LOS) path, neglecting real-world factors such as multipath effects and shadow fading, and is therefore often used as a theoretical reference. Its mathematical expression originates from the Friis transmission equation, with the basic form as follows:

$$P_r = P_t \frac{G_t G_r \lambda^2}{(4\pi R)^2} \quad (2.7)$$

where P_r and P_t denote the received and transmitted power respectively, G_t and G_r are the gains of the transmitting and receiving antennas, λ is the wavelength,

and R is the distance between transmitter and receiver. When converted to logarithmic form and expressed in decibels (dB), the equation becomes the commonly used Free Space Path Loss (FSPL) model:

$$PL(d) = 20\log_{10}(d) + 20\log_{10}(f) - 147.55 \quad (2.8)$$

In this equation, d represents the distance between the transmitter and receiver (in meters), and f is the signal frequency (in Hz). However, in real indoor environments, obstacles such as walls, furniture, and human bodies cause the signal to undergo reflection, diffraction, and scattering. These effects render the FSPL model insufficient for accurately describing practical path loss. Therefore, this study adopts the Log-distance Path Loss Model [6][16] to model the relationship between RSSI and distance. The model is mathematically expressed as:

$$PL(d) = PL(d_0) + 10n\log_{10}\left(\frac{d}{d_0}\right) + X_\sigma \quad (2.9)$$

where $PL(d)$ denotes the path loss at distance d (in dB), $PL(d_0)$ is the measured RSSI at a reference distance d_0 , and n is the path loss exponent, which varies depending on the environment. X_σ is a zero-mean Gaussian random variable used to model shadow fading; when shadowing is not considered, this term is set to zero. Additionally, this model can be reformulated in terms of the received signal strength (RSSI), leading to a simplified version such as Equation (1.1). This form is commonly used in practical indoor positioning systems, especially in scenarios where RSSI is used as the input for distance estimation and position inference.

Compared with the fixed path loss model, the log-distance model offers greater flexibility and environmental adaptability. It more accurately reflects signal attenuation

behavior within buildings and is widely applied in wireless sensing, indoor positioning, and network planning domains.

2.4.3 Filtering Methods

Due to the significant fluctuations and noise present in RSSI signals in indoor environments, directly using the raw RSSI values for distance estimation and positioning can lead to large errors. Therefore, smoothing the collected RSSI data before performing positioning calculations is an important step in improving positioning accuracy. This study adopts two common filtering methods: Moving Average Filtering [17] and Kalman Filtering [17][18], which are introduced as follows.

Moving Average Filtering is one of the simplest and most widely used smoothing methods. The principle is to take the average of the current value and the previous $n - 1$ data points as the current filtered output, smoothing out random fluctuations in the data. The mathematical expression is given by :

$$r_t = \frac{1}{n} \sum_{i=0}^{n-1} r_{t-i} \quad (2.10)$$

where r_t is the filtered RSSI value, r_{t-i} is the $t - i$ -th original data point, and n is the length of the moving window. This method is simple to implement and requires low computational cost. However, it has a slower response to changes in the signal and cannot effectively handle sudden changes or nonlinear noise in the data.

Kalman filtering is an optimal recursive filter designed to estimate the state of a dynamic system from noisy measurement data. Its strength lies in its ability to handle time-varying systems and to model noise and uncertainty. In the context of RSSI signal

smoothing, Kalman filtering treats the RSSI value as the state variable of a dynamic system. The filter operates in two main phases: the Prediction Phase and the Update Phase. In the Prediction Phase, the filter predicts the current state of the system and its associated uncertainty based on the best estimate from the previous time step. The equations are as follows:

$$\hat{x}'_k = A_k \hat{x}_{k-1} + B_k u_k \quad (2.11)$$

$$P'_k = A_k P_{k-1} A_k^T + Q_k \quad (2.12)$$

The term $\hat{x}'_k$ represents the a priori estimate, which is the predicted state at the current time step, while P'_k denotes the a priori error covariance, indicating the uncertainty of the prediction. The update phase utilizes the actual observation at the current time to correct the prediction. The corresponding formulas are as follows:

$$K_k = \frac{P'_k C_k^T}{C_k P'_k C_k^T + R_k} \quad (2.13)$$

$$\hat{x}_k = \hat{x}'_k + K_k (z_k - C_k \hat{x}'_k) \quad (2.14)$$

$$P_k = (I - K_k C_k) P'_k \quad (2.15)$$

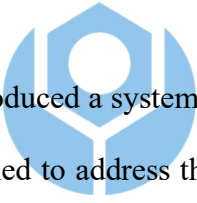
In these equations, K_k is the Kalman gain; $\hat{x}_k$ is the a posteriori estimate, representing the updated prediction of the next state; and P_k is the a posteriori error covariance, reflecting the uncertainty of the next predicted state. By continuously predicting the next RSSI value and correcting it based on new observations, the Kalman filter can effectively eliminate random noise in RSSI signals and significantly enhance their stability and accuracy.

2.5 Literature Review

This section presents a detailed review and analysis of recent research on indoor positioning based on RSSI. The studies are mainly categorized into two approaches: model-based methods and fingerprint-based methods.

2.5.1 Model-Based Indoor Positioning Studies

In 2021, L. S. De Oliveira et al. published a study in IEEE, proposing a low-cost indoor positioning system that combines multilateration and Kalman filtering [19]. The system was primarily designed for the navigation of autonomous mobile robots, using ESP32 microcontrollers to transmit and receive RSSI signals. Kalman filtering was applied to smooth out the position fluctuations caused by random variations in RSSI. Experiments conducted in a 9.7m x 13m environment achieved an average positioning error of 2.33 meters.



In 2023, Y. Assayag et al. introduced a system called ADAM in the IEEE Internet of Things Journal. This system aimed to address the limitation of traditional path loss models using fixed parameters across an entire environment, which often results in poor accuracy [20]. ADAM dynamically selects optimal anchor nodes and applies different parameters in the log-distance model to adapt to the varying signal propagation characteristics of different regions. Tested in a 45m x 16m educational building, the system achieved an average error of 2.93 meters without requiring prior on-site calibration or model training.

In 2024, Dariusz Janczak et al. proposed a novel indoor positioning method using Bluetooth Low Energy (BLE) and RSSI, published in IEEE Access [21]. The method was designed for personnel tracking in evacuation monitoring systems. To reduce computational complexity, the authors used an RSSI estimator in the form of a moving

average filter to improve distance estimation accuracy. Experiments were conducted with five receivers in an 11m x 23m area, resulting in an average positioning error of 2.5 meters.

In 2024, J. Li et al. presented a paper proposing a novel method to improve distance estimation accuracy in RSSI-based indoor positioning [22]. The approach involved averaging multiple high-frequency RSSI observations to refine the signal data. A wireless signal fading model was then fitted using various correlations between the averaged RSSI values and distance, aiming to obtain more accurate distance estimations. Experimental results showed that the method significantly enhanced ranging accuracy compared to existing approaches without requiring additional hardware.

In 2022, F. Serhan Danis et al. conducted a study focused on improving the performance of indoor positioning and tracking systems through real-time preprocessing of measured RSSI data [23]. The system was based on a hidden Markov model and employed a particle filter algorithm to estimate latent position variables. The study first compared three filtering techniques: mean, median, and maximum filters. Results showed that maximum filtering significantly improved performance in standard particle filter implementations. Further investigations explored various model parameters in both static and adaptive particle filter applications. Experiments demonstrated that maximum filter preprocessing could improve real-time positioning performance by more than 20%.

In 2022, L.S. de Oliveira et al. implemented a hybrid indoor positioning system (H-IPS) that combined two RSSI-based methods: multilateration (MLT) and fingerprinting (FP) [24]. Additionally, Kalman filtering (KF) was applied to reduce the

individual positioning errors of both techniques. To maximize performance, the study further performed track-to-track fusion (TTF) on the outputs of the two Kalman filters. Experimental results showed that H-IPS outperformed standalone fingerprinting by 21% and original multilateration by 52% in accuracy.

In 2022, B. Yang et al. proposed addressing indoor positioning issues using RSSI-based set-membership filtering [25]. First, they introduced a novel least squares curve fitting (LSCF) method to estimate transmission power and path loss exponent in RSSI-based positioning. Since trilateration is sensitive to noisy distance measurements, a convex optimization approach was developed to obtain state ellipsoid estimates. Subsequently, a recursive algorithm was constructed to compute global ellipsoids that guarantee the true target's location within each time step. Experimental results demonstrated the proposed method's accuracy and effectiveness in indoor positioning.

In 2021, Y.-B. Park et al. pointed out that trilateration, commonly used in indoor environments, is limited by the requirement that three circles intersect at a single point [26]. Many correction algorithms have been studied to address this issue. However, most existing methods apply different algorithms separately for cases where the three circles do not intersect at a point. They proposed a novel unified trilateration method that can be consistently applied in all scenarios, whether or not the three circles intersect at one point. The authors validated the algorithm's performance through Monte Carlo simulations.

In 2021, W. Njima et al. noted that indoor positioning can be achieved using a pairwise distance matrix among nodes, but such matrices are often incomplete and affected by distance estimation errors and RSSI measurement noise [27]. To address this,

the paper proposed a convolutional neural network (CNN) to denoise the completed distance matrix. Then, trilateration was applied to the recovered Euclidean distance matrix (EDM) to locate unknown nodes. The method was tested in simulation environments based on real propagation models, showing superior performance over classical approaches in both EDM recovery and positioning accuracy.

In 2021, Bruno Rodrigues et al. proposed BluePIL, a fully passive Bluetooth device identification and localization system [28]. The system is designed as a distributed streaming architecture capable of near real-time results and supports contact and activity tracking with varying data privacy levels. BluePIL identifies devices based on partial Bluetooth address information and employs a modified multilateration algorithm combined with a path loss model for device localization. Experimental results showed localization accuracy ranging from 1 to 1.4 meters in approximately 12 m² and 25 m² spaces.

In 2021, R.D. Ainul et al. presented an improved indoor positioning system that uses a Kalman filter (KF) to reduce inconsistencies in RSSI transmissions from Bluetooth Low Energy (BLE) wearable devices at each time unit [29]. They adopted the MultiQuad algorithm, which combines multilateration and quadratic weighting, as a deterministic position estimator. Results demonstrated that combining KF and MultiQuad improved position estimation accuracy by up to 90.32%, achieving a mean squared error (MSE) of 1.15 meters. This combined algorithm significantly reduced estimation errors compared to using traditional multilateration alone, which had errors up to 6.99 meters.

In 2020, Shi Y. et al. proposed a three-region RSSI classification and tracking algorithm as an RSSI filter [30]. The proposed filter showed favorable properties: the bias was less than 1 dBm when comparing a small number of RSSI test samples to large RSSI samples collected over about 10 minutes, and the subclassified RSSI followed a normal distribution when the minimum sample size exceeded 20. Experiments verified that the filter effectively improved path loss exponent fitting and position calculation. Positioning stability increased by 4.45 times based on sample standard deviation, positioning accuracy improved by 20.5%, and overall error was less than 1.46 meters.

2.5.2 Fingerprinting-based Indoor Positioning Studies

In 2024, T. Kosaka et al. introduced a novel method called the 2-Step Deep Neural Network, specifically designed for indoor positioning using Received Signal Strength Indicator (RSSI) data [31]. This approach aims to enhance estimation accuracy by using a single coordinate (either x or y) as a feature, representing an improvement over the previously proposed 2-Step Extreme Gradient Boosting. Key innovations include the transition from XGBoost to a deep neural network (DNN), along with optimized training data to develop a flexible learning model tailored to location coordinates. Comprehensive simulations demonstrated that the proposed two-step robust DNN achieved excellent estimation accuracy while maintaining an unconstrained dataset.

In 2023, P. E. Numan et al. proposed a DNN-based indoor positioning method that utilized WiFi Fine Time Measurement (FTM) and RSSI as environmental features to provide accurate location estimation [32]. Their method employed a DNN to process raw FTM and RSSI measurements for self-learning and generated enhanced ranging

information in the presence of measurement noise. Experimental data were collected from real-world scenarios using commercial devices in two different indoor office environments. The proposed solution was evaluated based on positioning mean squared error (MSE) and demonstrated superior accuracy compared to state-of-the-art methods.

In 2023, S. Deng et al. proposed a calibration-free Wi-Fi image fingerprinting indoor positioning method named Robust Radio Image Fingerprint Localization (RRIFLoc) [33]. First, Signal Strength Difference (SSD) fingerprints and RSSI kurtosis were derived from RSSI fingerprints to mitigate positioning accuracy degradation and weak interference resistance caused by fingerprint drift and feature disappearance. Next, RSSI, SSD, and kurtosis were fused to construct the Robust Radio Image Fingerprint (RRIF). Finally, the RRIFLoc model used the generated RRIF and a deep residual network for position estimation. Experimental results on public datasets showed that this method reduced the average positioning error by 56.87% compared to state-of-the-art indoor fingerprinting methods.

In 2023, Y. Li et al. proposed a novel hybrid measurement and simulation approach for generating fingerprint feature input sets to drive position prediction models [34]. Simulated feature parameters were incorporated as part of the neural network input, improving the original artificial neural network structure and expanding the sample feature set of the neural network inputs. Using an airport indoor arrival hall as a case study, static and dynamic positioning tests were conducted combining simulation and measurement fingerprint construction methods. Compared to conventional positioning based solely on fingerprint collection, the proposed method reduced static positioning error by 42.2% and dynamic positioning error by 46.9%.

In 2022, W. F. Hassen et al. proposed a WiFi- and convolutional neural network (CNN)-based indoor positioning system [35]. Their model used CNN to match RSSI images corresponding to reference points (RPs) with their locations. However, correlation among reference points caused RP ambiguity in RSSI images. To overcome this, the study adopted a distance-based label distribution (DBLD) method for RP encoding, which considers the distribution of neighboring RPs when encoding a single RP. To further improve performance, a random forest regression error correction model (ECM) was applied to adjust predicted target positions. Experimental results demonstrated high accuracy with an average positioning error of 0.72 meters.

In 2021, D. D. Nguyen et al. proposed a method combining machine learning (ML) and an improved k-nearest neighbors (kNN) algorithm [36]. The core idea used Gaussian Process Regression (GPR) for online position prediction, assisting the kNN algorithm in narrowing the search space to reduce computational cost. To evaluate confidence in estimated poses and optimize trajectories, the distribution of k nearest neighbors was analyzed. Experiments on the BBIL dataset containing noisy RSSI signals showed that this system significantly outperformed traditional kNN and weighted kNN (WkNN) algorithms. In a $10\text{ m} \times 25\text{ m}$ room, the optimized trajectory achieved a root mean square error (RMSE) of 1.78 meters, demonstrating competitive performance.

Chapter 3 Proposed Method

This chapter introduces the system architecture of the adaptive path loss model based on RSSI, including data collection, data processing, positioning framework and methods, as well as performance evaluation of the positioning model.

3.1 System Architecture

The system architecture of this study consists of four components: data collection, data processing, positioning framework and methods, and performance evaluation, as illustrated in Figure 3.1.

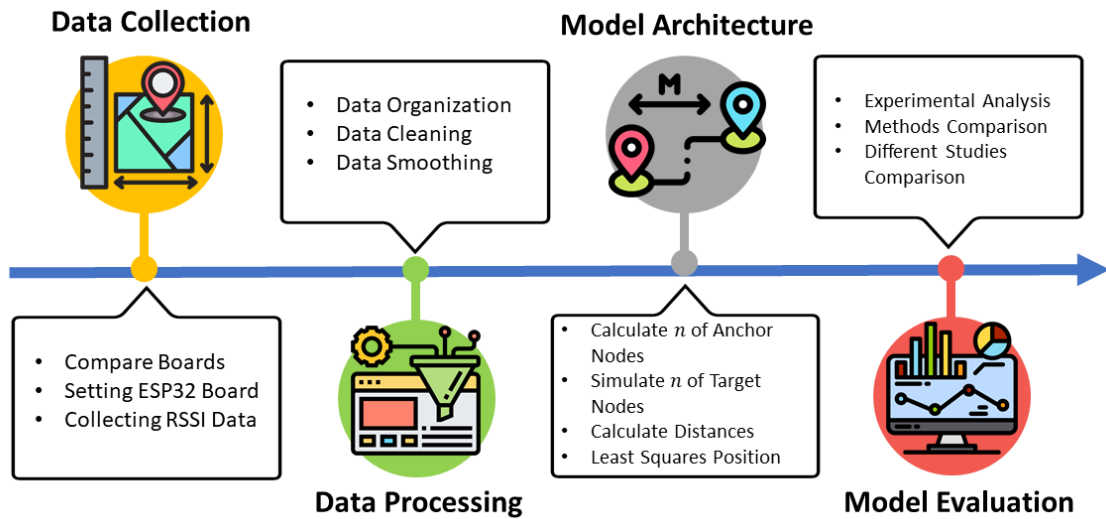


Figure 3.1 Proposed System Architecture

Data Collection Phase is responsible for acquiring the raw RSSI data. The main tasks include device selection and configuration to ensure the chosen microcontrollers meet the experimental requirements; ESP32 board settings covering communication parameter configurations; and collecting RSSI data by performing signal measurements at multiple locations within the planned experimental environment.

Data Processing Phase focuses on the preprocessing of raw RSSI data to improve data quality and stability. Detailed tasks include data organization, structuring the unorganized raw log data; data cleaning, removing outliers using statistical methods; and data smoothing, reducing signal fluctuations and noise through filtering algorithms.

Model Architecture Phase is the core of the positioning method in this study, responsible for converting the processed RSSI data into final position estimates. It consists of four key steps: calculating the path loss exponent (n value) for anchor nodes to estimate the environmental attenuation; simulating the target node's n value to achieve adaptive path loss; calculating the distance by converting RSSI into distance based on the adaptive n value; and least squares positioning, which uses optimization algorithms to determine the target position.

Model Evaluation Phase is responsible for comprehensively assessing the performance of the positioning system. The main tasks include experimental result analysis with statistical evaluation of positioning errors; method comparison, benchmarking the proposed method against other common indoor positioning techniques; and related work comparison, analyzing the strengths and weaknesses of this study relative to existing literature.

3.2 Data Collection

This section introduces the selection and configuration of equipment for data collection, the experimental environment where the dataset was collected, and the content of the dataset.

3.2.1 Device Selection and Configuration

To conduct RSSI-based indoor positioning experiments, this study initially compared common microcontrollers available on the market that possess wireless communication capabilities. These included NodeMCU-32S ESP32[37], Arduino ESP8266[38], and Arduino ESP32[39]. After comprehensive consideration of processing performance, GPIO count, wireless module functionality, development convenience, and experimental stability, the Arduino ESP32 module, which supports 2.4 GHz Wi-Fi and Bluetooth 4.2 BLE, was ultimately selected as the experimental device.

Prior to the formal experiment, preliminary tests were conducted in a classroom measuring 12 meters $\times$ 5 meters. The purpose of these tests was to compare the performance of ESP32's built-in Wi-Fi and BLE modules in terms of RSSI signal strength and stability. In this preliminary experiment, one ESP32 device served as the signal transmitter, broadcasting signals in both Wi-Fi and BLE modes, while a mobile phone receiver collected RSSI data at various distances.

The experimental results, as shown in Figure 3.2, indicated that the RSSI value fluctuations of the ESP32's Wi-Fi module were significantly lower than those of the Bluetooth module under 0.2m and 13m distance conditions. Furthermore, the Wi-Fi module demonstrated overall more stable signal strength. Consequently, the Wi-Fi module of ESP32 was adopted for subsequent positioning experiments as the RSSI data transceiver for this research.



Figure 3.2 RSSI Stability Comparison of Wi-Fi and BLE at Short and Long Distances. (a) Wi-Fi Short Range. (b) Wi-Fi Long Range. (c) BLE Short Range. (d) BLE Long Range.

This study configured the ESP32 devices to simultaneously act as both a Wi-Fi Access Point (AP) and a Station (STA). During the initialization phase, the ESP32 activates an AP named with its unique ID, allowing other devices to connect. In operation, the ESP32 periodically performs non-blocking Wi-Fi scans to actively detect surrounding Wi-Fi networks. To ensure rapid signal acquisition, the code specifically set the scan dwell time for each Wi-Fi channel to 20 milliseconds. This optimization allowed each complete Wi-Fi scan, including the collection of signals from 13 Wi-Fi channels and the acquisition of RSSI data from various APs, to be completed in approximately 260 milliseconds.

3.2.2 Experimental Environment

The experimental environment for this study was located in the main hall on the first floor of the St. Bonaventure Hall at Fu Jen Catholic University, covering an area of approximately 15 meters $\times$ 11 meters. Within this space, four ESP32 devices were

strategically placed at the four corners of the hall to serve as anchor nodes for the system, as shown in Figure 3.3(b). An additional ESP32 device functioned as a simulated mobile target node, used for collecting RSSI data at various locations.



Figure 3.3 Experimental environment. (a) Layout of the environment and interference objects. (b)

Anchor node deployment.

Due to the susceptibility of RSSI signals to environmental factors such as human movement, weather changes, and fluctuations in temperature and humidity, this study distributed data collection across three different dates. This was deliberately done to enhance the representativeness and coverage of the experimental data, by encompassing multiple time slots within a day. During these periods, weather conditions also varied, including sunny, cloudy, and rainy days. Furthermore, all experiments were conducted on normal school days, allowing for the natural ingress and egress of faculty and students in the hall. This approach helped to simulate the RSSI variations expected in actual operating environments.

As depicted in Figure 3.3(a), the experimental environment also contained major obstacles that could interfere with signals. These included pillars within the space, metal light tracks on the ceiling, and iron doors inside the hall. Such environmental interferences are capable of causing multipath effects and signal attenuation, thereby impacting the stability of RSSI signals.

3.2.3 Data size

This study established 20 positioning points as experimental sampling coordinates, with their precise locations depicted in Figure 3.4 (please replace with your actual figure number). These points collectively covered the entire experimental environment. At each positioning point, RSSI data was collected, yielding an average of approximately 3,500 RSSI signal values per point.

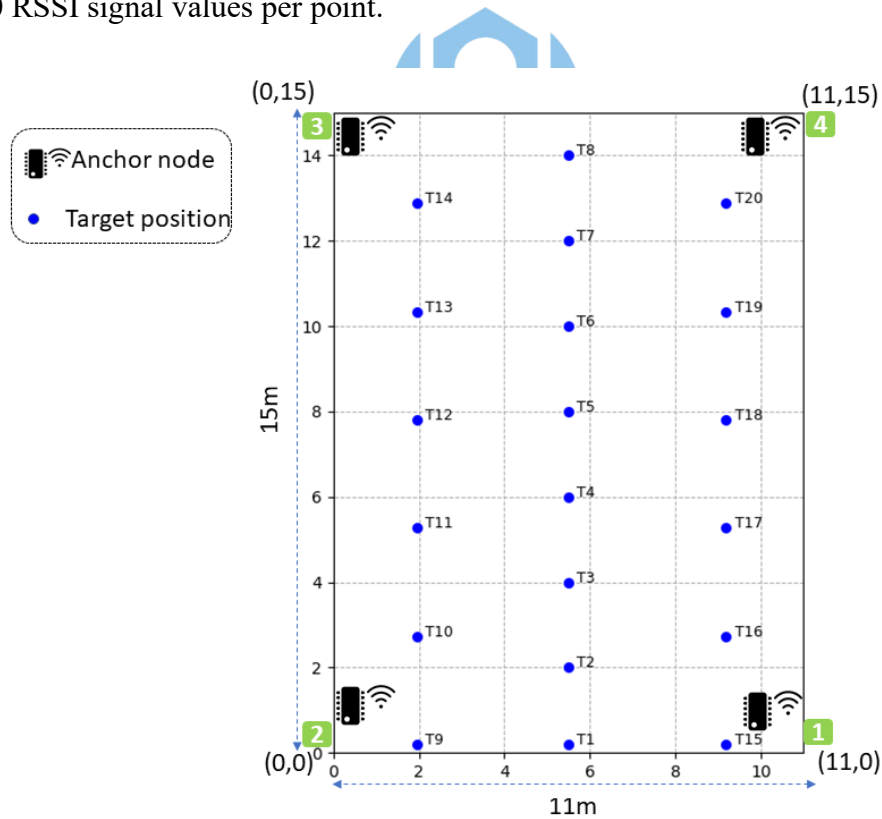


Figure 3.4 Experimental Sampling Coordinates of the 20 Target Points.

Due to the system configuration of 4 ESP32 anchor nodes and 1 target node, with each node possessing Wi-Fi sending and receiving capabilities, multiple pairs of interconnected devices could form mutually transmitting and receiving signal connections during the experiment. This study employed a full-duplex transceiver mode, meaning each pair of devices recorded each other's received RSSI values. Therefore, each positioning point generated:

$5 \text{ device transceiver combinations} \times 3,500 \text{ data points} \times 20 \text{ positioning points} =$ a total of approximately 350,000 RSSI data points.

The aforementioned data encompasses signal strength variations across different distances and directions among all nodes, providing sufficient samples for subsequent RSSI model training, path loss parameter estimation, and positioning algorithm validation.

3.3 Data Processing



After signal collection, this study performed a series of preprocessing steps on the raw RSSI data to ensure data quality and stability, providing reliable input for subsequent positioning model calculations. The data processing workflow primarily includes three stages: data organization and classification, outlier removal, and signal smoothing.

Firstly, the data organization and classification stage involved structuring the collected RSSI signals. Since the experimental data covered signals detected by different receiving nodes from various transmitting nodes at different times, raw log file data needed to be parsed and classified. We extracted timestamps, device names, and corresponding RSSI values from the raw files. This data was then used to generate independent Excel files based on its measurement time intervals and communication links.

During the Excel file generation, data trimming was also performed to ensure that each file had the same data length, facilitating subsequent batch processing and analysis.

Secondly, outlier removal is crucial for enhancing the purity of RSSI data. Due to the complexity of indoor environments, RSSI signals are susceptible to instantaneous interference, which can generate extreme outlier values. This study adopted the Inter-quartile Range (*IQR*) method to identify and remove these outliers. Specifically, for every 100 RSSI samples as a processing unit (chunk), the first quartile ($Q1$) and third quartile ($Q3$) were calculated for each data block. Upper and lower bounds were then set at $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ respectively. Data points falling outside these ranges were considered outliers and were discarded, thereby effectively removing anomalous values from the dataset.

Finally, after outlier removal, the RSSI data undergoes signal smoothing to reduce the impact of random noise and enhance signal stability. This study primarily compared two methods: the Kalman Filter and the Moving Average Filter. Their mathematical foundations and principles have been detailed in Section 2.2.3.

The Moving Average Filter is a simple smoothing technique that achieves its effect by averaging values within a fixed time window, requiring no complex models. To evaluate its performance, we set different window sizes in our experiments, including 5, 20, 50, and 100. The Kalman Filter, on the other hand, is an optimal recursive filter based on a statistical model. It operates through two phases—prediction and update—dynamically adjusting estimated values based on changes in system states over time. In implementation, we set the initial estimate $\hat{X}_0 = X_0$ and initial error covariance $P_0 = 1$, and adopted common parameters empirically verified as suitable for RSSI data: process

noise covariance $Q = 1 \times 10^{-5}$ and measurement noise covariance $R = 0.01$. The effects of both filtering methods are shown in Figure 3.5.

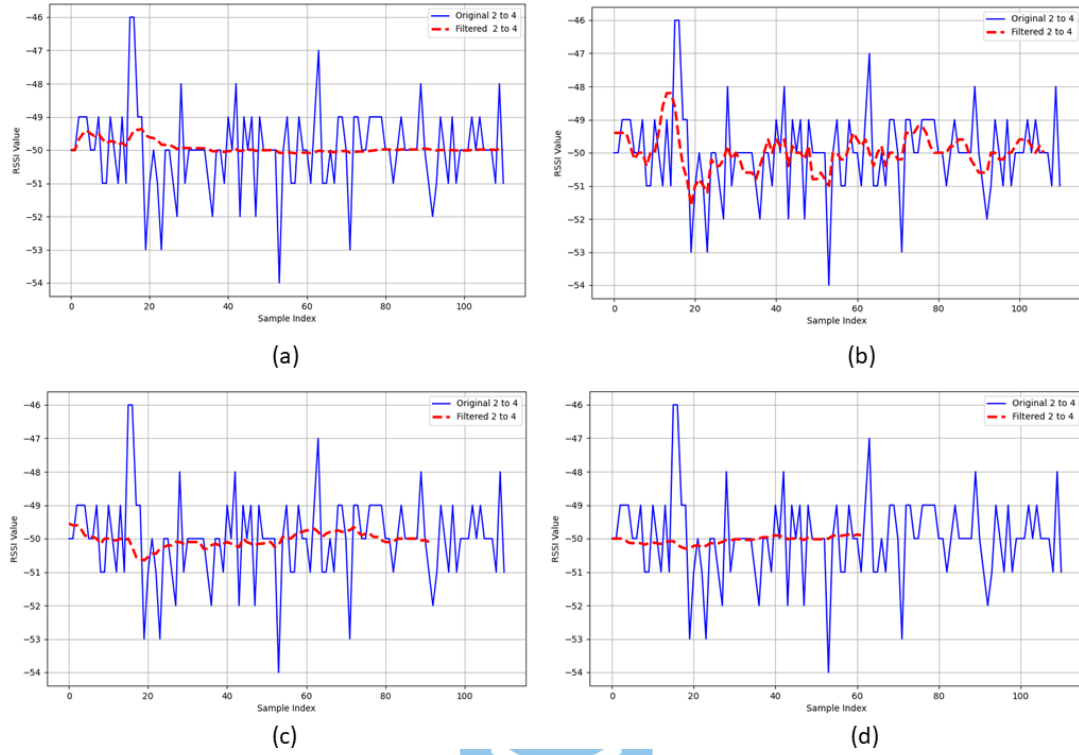


Figure 3.5 Comparison of Filtering Effects. (a) Kalman filter. (b) Moving average filter (window size = 5). (c) Moving average filter (window size = 20). (d) Moving average filter (window size = 100).

To objectively evaluate the performance of the two filtering methods, we conducted data tests at three specific locations (T1, T3, T6) within the experimental environment. The RSSI values, processed by both filtering methods, were then used in the path loss model to calculate the distances from the target node to the anchor nodes, as shown in Table 3.1(a). The average accuracy of the four sides was also calculated, as presented in Table 3.1(b). Experimental results indicated that although the accuracy of the Moving Average Filter at location T3 was slightly higher than that of the Kalman Filter when the window size was set to 100, the Kalman Filter generally outperformed the Moving Average Filter in overall performance. Therefore, this study ultimately

selected the Kalman Filter as the primary method for smoothing RSSI signals, with the processed data serving as input for subsequent positioning model calculations.

Table 3.1 Filtering Performance Comparison at T1, T3, and T6. (a) Estimated Distances. (b) Average Accuracy of distances.

TN	Item	Real d(m)	KF d(m)	MAF_5 d(m)	MAF_20 d(m)	MAF_50 d(m)	MAF_100 d(m)
T1	d1	5.5	5.14	7.44	5.66	5.6	5.68
	d2	15.97	12.81	9.13	10.27	10.26	10.26
	d3	15.97	13.02	6.71	13.03	12.96	12.9
	d4	5.5	1.70	2.30	2.22	2.22	2.23
T3	d1	6.8	3.5	2.87	3.39	3.39	3.4
	d2	8.9	5.07	4.23	4.55	4.56	4.55
	d3	8.9	8.7	6.71	9.89	9.89	9.97
	d4	6.8	4.87	3.66	4.37	4.29	5.18
T6	d1	11.4	9.30	9.49	9.50	9.49	9.54
	d2	5.6	1.95	2.27	2.26	2.25	2.24
	d3	5.6	6.21	7.79	7.76	7.64	7.57
	d4	11.4	4.99	5.25	5.25	5.23	5.22

(a)

TN	MAF_5(%)	MAF_20(%)	MAF_50(%)	MAF_100(%)	KF (%)
T1	51.43	70.84	70.99	70.57	71.53
T3	58.56	63.53	63.26	66.32	62.15
T6	57.68	57.79	58.26	58.57	62.32

(b)

3.4 Proposed Positioning Method

This section will elaborate on the core positioning method proposed in this study. The core positioning method aims to enhance the indoor positioning accuracy based on RSSI by utilizing an adaptive path loss model. Given the limitations of traditional path loss models, where the path loss exponent n is fixed or requires extensive calibration, this method's core innovation lies in enabling the model to adaptively fit the n value for different communication links and environmental conditions. As shown in Figure 3.6 (please replace with your actual figure number), the architecture of this study's positioning method comprises four main steps. All RSSI data, after being cleaned and processed, will sequentially be fed into this model for computation.

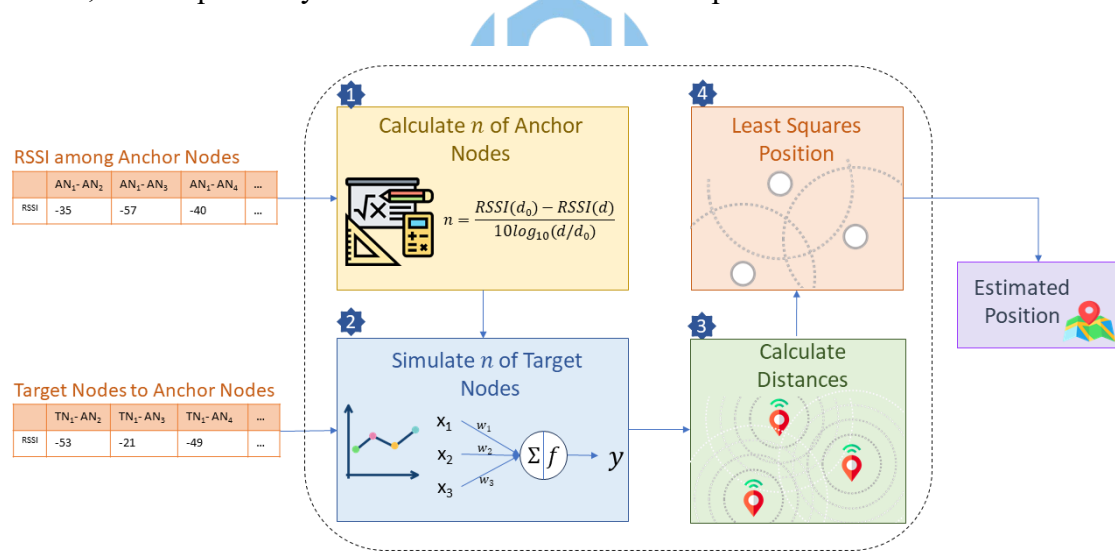


Figure 3.6 Architecture of the Proposed Method.

The four computational steps of this method are sequentially as follows:

Calculating n for Anchor Nodes : Utilizes known RSSI signal values and true distances between anchor nodes to calculate the actual path loss exponent n in the environment.

Simulating n for Target Nodes : Simulates the target node's signal values using the RSSI signals from anchor nodes, and applies the derived weights to the anchor nodes' true n values to simulate the n values from the target node to each anchor node.

Calculating Distances : Once the adaptive n values from the target node to each anchor node are obtained, the distance from the target node to each anchor node can be calculated via the path loss model.

Least Squares Position Estimation : Finally, the Least Squares positioning algorithm is employed to estimate the final coordinates of the target node.

The following subsections will elaborate on these core steps.

3.4.1 Adaptive Path Loss Model

This section will elaborate on the core innovation of this study—the adaptive path loss model.

As described in Section 2.2.2, the Log-distance Path Loss Model serves as the foundation for distance estimation based on RSSI. Its general form is given by Equation (1.1). In this model, the path loss exponent n is a critical parameter reflecting environmental attenuation characteristics, and its value varies significantly across different environments. However, in traditional methods, the n value is typically set as a fixed empirical value or determined through extensive preliminary calibration. This static or pre-calibrated n value often fails to accurately reflect the dynamic changes in signal

propagation characteristics within complex indoor environments, such as variations caused by human movement, furniture rearrangement, or humidity fluctuations. Given the logarithmic relationship between RSSI and distance, even a minor change in the n value can lead to distance estimation errors of several to tens of times. This precisely constitutes a key bottleneck leading to insufficient positioning accuracy based on RSSI, and it is also the primary motivation for this study's adaptive model. The core workflow of this adaptive model is shown in Figure 3.6, primarily divided into four steps, with the first two steps focusing on the adaptive estimation of the n value.

The core of this approach lies in the fact that the adaptive path loss exponent n is not based on a fixed empirical value, but is dynamically estimated using real-time data from known Anchor Nodes (ANs). By modeling and inferring the path loss exponent for the Target Node (TN) through the RSSI relationships among the ANs, the system can automatically adapt to environmental changes and derive the most suitable n value for the current surroundings. This ensures that distance estimation more accurately reflects the subtle variations in wireless signal propagation characteristics within complex indoor environments.

3.4.2 Calculating n for Anchor Nodes

The objective of this step is to utilize the known RSSI signal values between anchor nodes and their true distances to calculate the actual path loss exponent n in the environment. Since we have 4 anchor nodes, they can transmit signals to each other,

forming multiple sets of known communication links, as illustrated in Figure 3.7

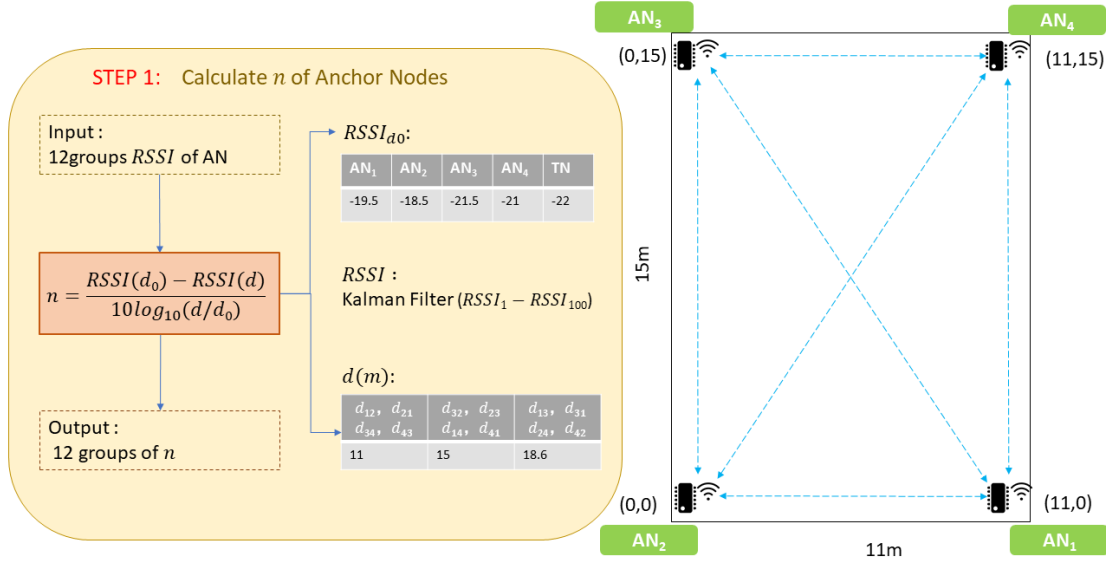


Figure 3.7 Flowchart of Calculating n for Anchor Nodes.

For instance, anchor node AN₁ can receive signals from AN₂, AN₃, and AN₄, thereby generating 3 sets of n values. Similarly, each anchor node is capable of receiving signals from the other 3 anchor nodes, resulting in a total of $4 \times 3 = 12$ sets of n values among the anchor nodes. Starting from the Log-distance Path Loss Model (Equation 1.1), we can transform it into a form for solving n :

$$n = \frac{RSSI_{d_0} - RSSI_d + X_\sigma}{10 \log_{10} \left(\frac{d}{d_0} \right)} \quad (3.1)$$

In actual calculations, since the distance d between anchor nodes is known, and $RSSI_{d_0}$ (at a reference distance of 1 meter) has been measured in an interference-free laboratory, and furthermore, the noise in the $RSSI$ values has been significantly reduced after Kalman filter processing, we can temporarily set X_σ to 0. Therefore, the n value between anchor nodes can be estimated using the following simplified formula

$$n_{AN_i \leftarrow AN_j} = \frac{RSSI_{d0} - RSSI_{AN_i \leftarrow AN_j}}{10 \log_{10} (d_{AN_i \leftrightarrow AN_j})} \quad (3.2)$$

Here, $n_{AN_i \leftarrow AN_j}$ represents the unidirectional path loss exponent when a signal is transmitted from anchor node AN_j to anchor node AN_i . $RSSI_{AN_i \leftarrow AN_j}$ indicates the RSSI value received by anchor node AN_i from anchor node AN_j . $RSSI_{d0}$ is the reference RSSI value measured at a reference distance $d0$ (1 meter). $d_{AN_i \leftrightarrow AN_j}$ denotes the physical distance between anchor node AN_i and AN_j . Through this method, we are able to calculate an empirical n value for each pair of anchor node communication links, representing the characteristics of their respective areas. In total, there are 12 sets of these n values.

3.4.3 Simulating n for Target Nodes

In the first step, we have calculated 12 sets of path loss exponents $n_{AN_i \leftarrow AN_j}$ for anchor nodes in the real environment. The objective of this step is to utilize these known n values between anchor nodes to infer the path loss exponent $n_{AN_i \leftarrow TN}$ from the target node to each of the 4 anchor nodes. This is crucial for realizing the adaptiveness of this research, aiming to enable the model to dynamically adjust the n value for the communication link based on the target node's immediate environment. A schematic diagram of this process is shown in Figure 3.8.

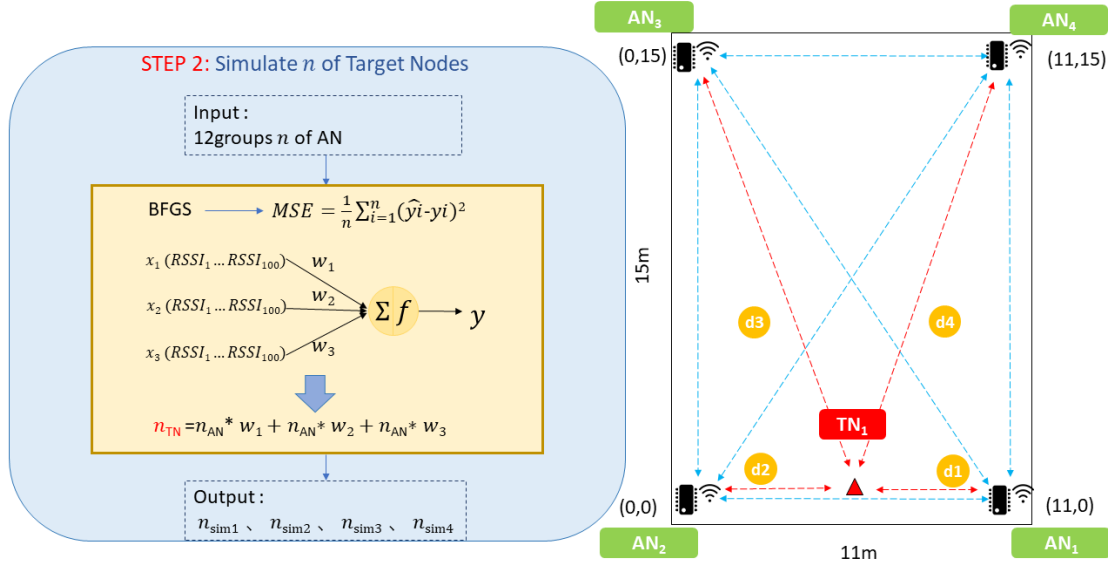


Figure 3.8 Flowchart of Simulating n for Target Nodes.

Based on the observed signal propagation patterns, this study infers the path loss exponent $n_{AN_i \leftarrow TN}$ from the target node to each anchor node. This value can be obtained by a weighted combination of environmental characteristics reflected by the RSSI values measured between anchor nodes. Specifically, for a given anchor node AN_i , we use the RSSI values it receives from other anchor nodes as a feature vector to infer the RSSI value that the target node would transmit to that anchor node.

Numerical optimization methods are employed here to estimate the best weighted combination of input variables to the target variable, with the objective of minimizing the Mean Squared Error (MSE). We define the RSSI values sent by anchor node AN_j to anchor node AN_i as the feature vector X (i.e., $X = [RSSI_{AN_i \leftarrow AN_1}, RSSI_{AN_i \leftarrow AN_2}, \dots]$). The RSSI signal from the target node to anchor node AN_i is defined as the target vector Y (i.e., $Y = RSSI_{AN_i \leftarrow TN}$).

To find the optimal weight combination w , we utilize the BFGS (Broyden–Fletcher–Goldfarb–Shanno) algorithm. BFGS is an unconstrained optimization algorithm that accelerates convergence by approximating the second derivative (Hessian matrix) of the objective function, thereby minimizing the loss function[40]. The form of the loss function is typically:

$$L(w) = \frac{1}{M} \sum_{k=1}^M \left(Y_k - \hat{Y}_k(w, X_k) \right)^2 \quad (3.3)$$

Here, $\hat{Y}_k(w, X_k)$ is the RSSI value predicted based on weights w and features X_k .

Through the BFGS algorithm, we obtain the optimal weights $w = [w_1, w_2, w_3]$. These weights represent how communication characteristics between different anchor nodes influence the target node's signal. Subsequently, these optimal weights are applied to the path loss exponents calculated in the first step, specifically to $n_{AN_i \leftarrow AN_j}$, to infer the n value from the target node to each anchor node:

$$n_{AN_i \leftarrow TN} = w_1 \cdot n_{AN_i \leftarrow AN_j} + w_2 \cdot n_{AN_i \leftarrow AN_j} + w_3 \cdot n_{AN_i \leftarrow AN_j} \quad (3.4)$$

By this method, we ultimately obtain four adaptive path loss exponents, $n_{sim1}, n_{sim2}, n_{sim3}, n_{sim4}$, for the target node to each of the four anchor nodes. These n values can better reflect the local environmental attenuation characteristics at the target node's location, thereby improving the accuracy of distance estimation.

3.4.4 Calculating Distances

Once the adaptive path loss exponent $n_{AN_i \leftarrow TN}$ for each anchor node is obtained, it can be applied to the Log-distance Path Loss Model to inversely derive the distance

from the target node to each anchor node, as illustrated in Figure 3.9. Based on the Log-distance Path Loss Model from Equation (1.1), and by again setting the reference distance d_0 to 1 meter and $X\sigma$ to 0, the formula can be transformed to solve for distance d :

$$d = d_0 \cdot 10^{\frac{RSSI_{d_0} - RSSI_d}{10n}} \quad (3.5)$$

Therefore, the distance $d_{AN_i \leftrightarrow TN}$ from the target node TN to anchor node AN_i can be estimated as:

$$d_{AN_i \leftrightarrow TN} = 10^{\left(\frac{RSSI_{d_0} - RSSI_{AN_i \leftrightarrow TN}}{10n_{AN_i \leftrightarrow TN}} \right)} \quad (3.6)$$

Where $d_{AN_i \leftrightarrow TN}$ is the estimated distance between the target node TN and anchor node AN_i . $RSSI_{d_0}$ is the reference RSSI value measured at a reference distance of 1 meter. $RSSI_{AN_i \leftrightarrow TN}$ is the RSSI value received by anchor node AN_i from the target node TN, which has been smoothed by the Kalman Filter as described in Section 3.3. $n_{AN_i \leftrightarrow TN}$ is the adaptive path loss exponent from the target node to anchor node AN_i , inferred as described in Section 3.4.1.2.

Through the aforementioned formula, we are able to obtain four estimated distance values from the target node to the four anchor nodes. These distance values will serve as input for the subsequent position calculation.

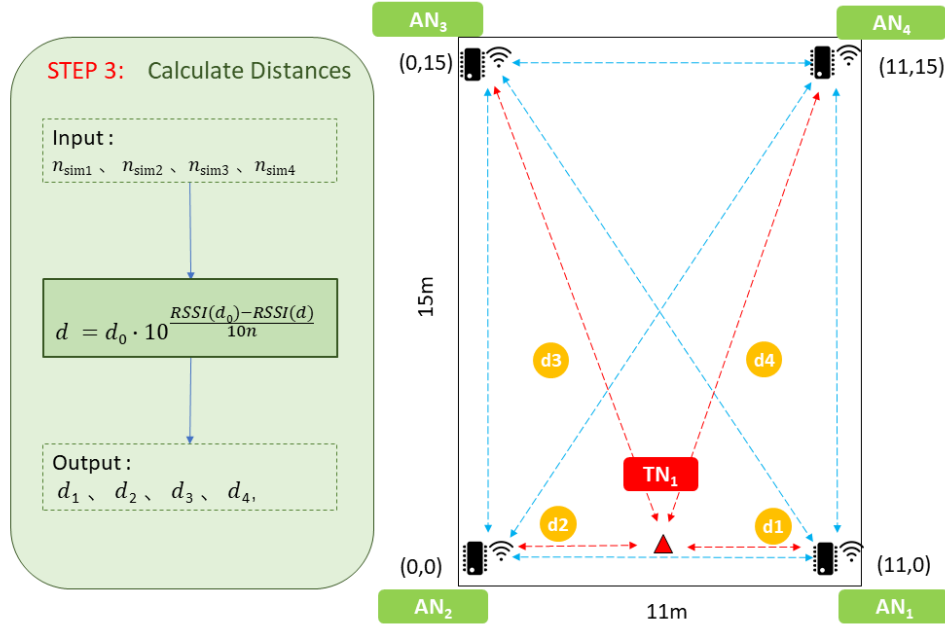


Figure 3.9 Flowchart of Calculating Distances.

3.4.5 Position Estimation

After obtaining the estimated distance values from the target node to multiple anchor nodes, due to inevitable errors in RSSI signal measurement and distance estimation processes, the circles formed by multiple anchor nodes may not intersect precisely at a single point. As shown in Figure 3.10, in an ideal scenario, the target node $T(x, y)$ should be located at the precise intersection point of all circles, as depicted in Figure 3.10(a). However, real-world situations are more likely to present as circles with a small overlapping region, as in Figure 3.10(b), or even no common intersection point, as in Figure 3.10(c). To address this issue, this study employs the Least Squares (LS) method [41] to determine the optimal estimated position of the target node.

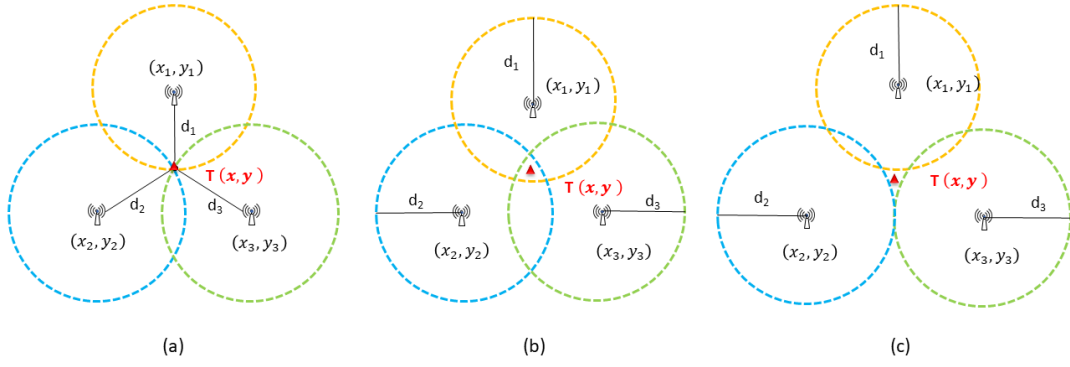


Figure 3.10 Various Trilateration Intersection Scenarios. (a) Ideal intersection. (b) Partial intersection. (c) No intersection.

The Least Squares (LS) method is a widely applied mathematical optimization technique aimed at estimating model parameters by minimizing the sum of squared errors between observed values and model-predicted values. In localization problems, its objective is to find the optimal position for the target point (x, y) such that the sum of squared errors between its estimated distances from all anchor points and its actual geometric distances to them is minimized.

Assuming the unknown coordinates of the target node are (x, y) , and there are N anchor nodes, with the i -th anchor node having known coordinates (x_i, y_i) . The estimated distance from the target node to anchor node i , obtained through Section 3.4.1, is d_i . The geometric distance from the target node to anchor node i is $d_i = \sqrt{(x - x_i)^2 + (y - y_i)^2}$. The distance relationships can be expressed as a series of non-linear equations:

$$\begin{cases} (x_1 - x)^2 + (y_1 - y)^2 = d_1^2 \\ \dots \\ (x_i - x)^2 + (y_i - y)^2 = d_i^2 \end{cases} \quad (3.7)$$

To apply the linear least squares method for solving, the above nonlinear system of equations typically requires linearization. This can be achieved by expanding and subtracting the distance equations. By subtracting the $i - th$ equation from the first equation, the x^2 and y^2 terms can be eliminated, yielding a series of linear equations. After reorganization, the system can be transformed into the form $AX = b$, where X is the unknown target node coordinate vector to be solved for, and matrix A and vector b are derived from the anchor node coordinates (x_i, y_i) and estimated distances, typically having the following forms:

$$A = \begin{pmatrix} 2(x_1 - x_i) & 2(y_1 - y_i) \\ 2(x_{i-1} - x_i) & 2(y_{i-1} - y_i) \end{pmatrix}, X = \begin{pmatrix} x \\ y \end{pmatrix} \quad (3.8)$$

$$b = \begin{pmatrix} x_1^2 - x_i^2 + y_1^2 - y_i^2 + d_1^2 - d_i^2 \\ \dots \\ x_{i-1}^2 - x_i^2 + y_{i-1}^2 - y_i^2 + d_{i-1}^2 - d_i^2 \end{pmatrix} \quad (3.9)$$

The error can be expressed as $e(X) = \|b - AX\|^2$. The objective of the Least Squares method is to minimize the objective function $e(X)$. The derivative of $e(X)$ is:

$$\frac{de(x)}{dx} = 2AA^T x - 2Ab = 0 \quad (3.10)$$

Therefore, we can obtain the position of the target node :

$$X = (AA^T)^{-1}Ab \quad (3.11)$$

Chapter 4 Performance Analysis

This chapter comprehensively evaluates the performance of the RSSI-based adaptive path loss model indoor positioning system proposed in this study. The content includes an introduction to evaluation metrics, analysis of positioning results and errors in actual experimental settings, a comparative analysis of this method against other positioning methods, and a comparison with related research. Through these analyses, this chapter aims to validate the performance of the proposed adaptive model.

4.1 Positioning Error Evaluation Metrics

This section aims to introduce the evaluation metrics used in this research. Through these evaluation metrics, the performance of the indoor positioning system proposed in this study can be objectively assessed.

4.1.1 Mean Absolute Error

In this study, Mean Absolute Error (MAE) is used to evaluate the average absolute deviation between the predicted and actual positions along each coordinate axis. This metric separately calculates the absolute errors on the X-axis and Y-axis, then takes their average. Such an evaluation provides a more detailed reflection of the model's prediction accuracy in each dimension. First, compute the absolute error on the X-axis and Y-axis for each test point. The calculation is as follows:

$$mae_{x_i} = |x_i - \hat{x}_i| \quad mae_{y_i} = |y_i - \hat{y}_i| \quad (4.1)$$

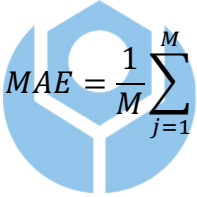
Next, calculate the mean point error for each test point :

$$mean_{error_i} = \frac{mae_{x_i} + mae_{y_i}}{2} \quad (4.2)$$

Then, compute the MAE for all test points within a single test batch :

$$MAE_{batch} = \frac{1}{N} \sum_{i=1}^N mean_{error_i} \quad (4.3)$$

Where N is the total number of test points in that test batch, (x_i, y_i) are the actual coordinates of the $i - th$ test point, and $(\hat{x}_i, \hat{y}_i)$ are the estimated coordinates of the $i - th$ test point. Furthermore, to evaluate the system's stability and generalization capability under different time or environmental conditions, this study further calculated the Total Mean Absolute Error (Total MAE). This Total MAE represents the average of the MAE results from all test batches. Its formula is expressed as:



$$Total\ MAE = \frac{1}{M} \sum_{j=1}^M MAE_j \quad (4.4)$$

Where M is the total number of test batches, and MAE_j is the Mean Absolute Error calculated for the $j - th$ test batch. The unit of MAE is typically meters (m).

4.1.2 Cumulative Distribution Function

The Cumulative Distribution Function (CDF) plot is an intuitive and comprehensive method for evaluating positioning errors. It illustrates the percentage of test points with positioning errors less than or equal to a specific threshold. In a CDF plot, the closer the curve is to the upper-left corner, the higher the positioning accuracy—indicating a larger proportion of points fall within a smaller error range. CDF plots are particularly useful for analyzing the following key metrics:

50% Error (P50, Median Error): This indicates that 50% of the positioning errors are less than or equal to this value. As the median of the positioning errors, it is less affected by outliers compared to the Mean Absolute Error (MAE), and thus better reflects the typical performance of the system.

90% Error (P90): This indicates that 90% of the positioning errors are less than or equal to this value. It represents the worst-case accuracy level achievable in most scenarios, making it a crucial indicator for applications with high stability requirements, such as emergency response and asset tracking.

4.2 Experimental Analysis

This section presents the experimental results of the proposed positioning method tested on 20 different target locations.

In the actual tests, we examined the impact of using different quantities of RSSI values—specifically 30, 100, and 500—on positioning performance. Since the total amount of collected RSSI data was fixed, the number of measurements per location varied depending on the selected number of RSSI values. The time required for a single positioning operation was almost entirely determined by the number of RSSI values used. This is because, when using the ESP32's Wi-Fi module, each scan takes approximately 260 milliseconds. As a result, the positioning time varied significantly based on the number of RSSI values. Table 4.1 summarizes the positioning performance comparison using different quantities of RSSI values.

Table 4.1 Performance Metrics for Varying RSSI Sample Sizes.

	RSSI_30	RSSI_100	RSSI_500
Total MAE	2.35m	2.33m	2.28m
Measurements	23	7	1
Execution Time	7.8s	26s	130s

From the table, it can be observed that as the number of RSSI signals increases, the overall MAE slightly decreases, but the execution time significantly increases. Considering the trade-off between positioning performance and time cost, this experiment ultimately adopted 100 RSSI signal values as the evaluation standard for all subsequent analyses. Under this evaluation standard, the test results for 20 target locations showed that the average absolute error (MAE) from 7 measurements per location was 2.33 meters, and the time required for one localization instance was approximately 30 milliseconds.

Figure 4.1 presents the overall MAE heatmap for the 20 target locations, visually illustrating the distribution of positioning errors at each point. Concurrently, Figure 4.2 displays the specific error data for each test point, sorted by error magnitude. This section will proceed with an in-depth analysis focusing on the four best-performing points and the four worst-performing points.

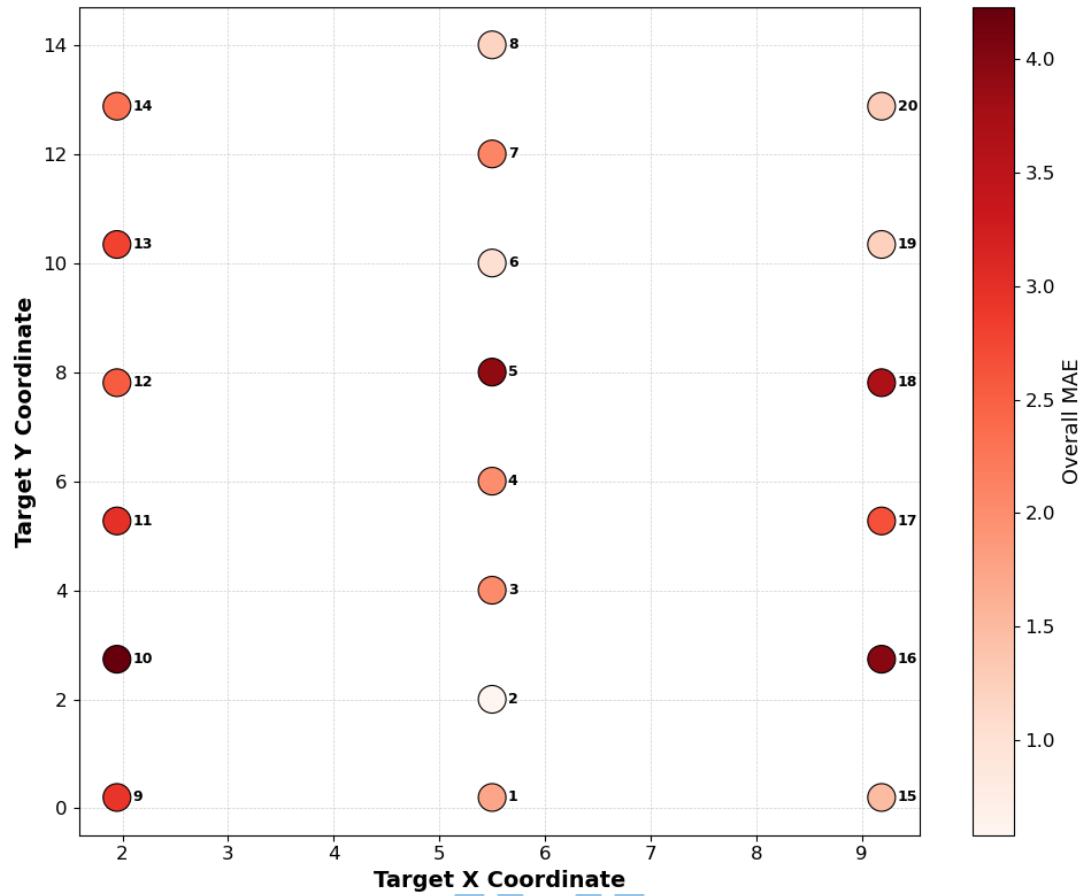


Figure 4.1 Total MAE Heatmap at 20 Target Locations

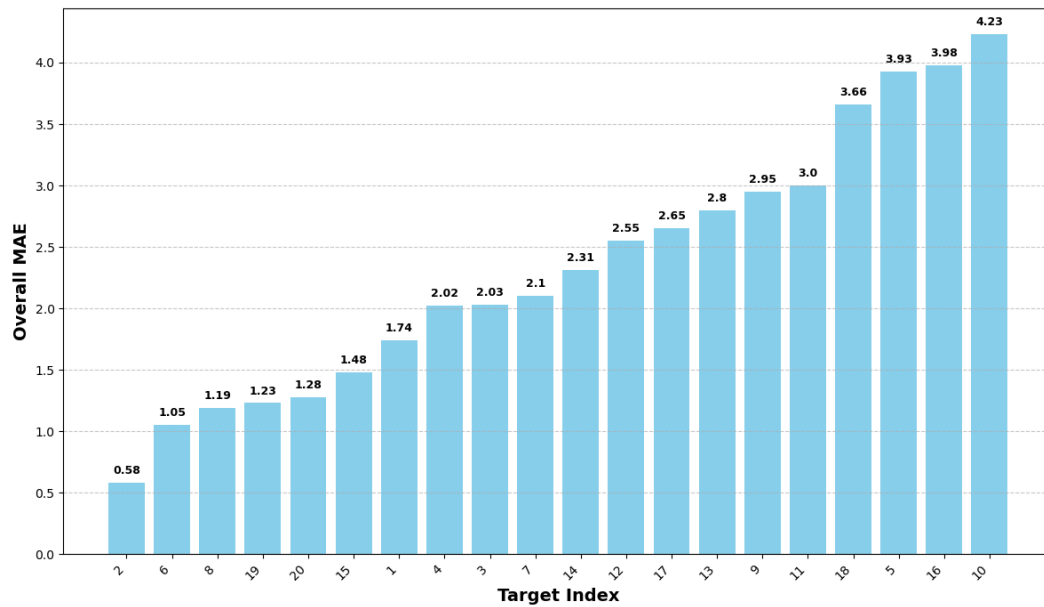


Figure 4.2 Total MAE for Each Target Point (Sorted)

Table 4.2 presents the estimated path loss exponent (n) and distance, along with the positioning error analysis, for some representative test points. Table 4.2(a) displays the test points with better positioning performance. From the data, it can be observed that for these well-performing points, the path loss exponent n in the model was well-fitted, with an average accuracy exceeding 85%. Consequently, the distances calculated based on these accurately simulated n values were also relatively accurate. This indicates that improving the accuracy of the path loss exponent n indeed helps to enhance distance accuracy, thereby improving the overall positioning accuracy.

Table 4.2(b) showcases the test points with the worst positioning performance. The numbers highlighted in blue indicate the communication links (sides) with the least accurate simulations for each point. We identified two categories of problems: first, errors caused by environmental interference, and second, issues of near-range instability due to algorithmic limitations. The inaccurate estimations for T5's d4 and T18's d3 can likely be attributed to the influence of environmental obstacles. T5's location is near the middle of two pillars in the experimental area. T18's location is adjacent to two iron doors. These elements lead to non-line-of-sight (NLOS) and shadowing effects, making it difficult for the model to accurately capture signal propagation characteristics.

A peculiar phenomenon was observed with T10's d2 and T16's d1: according to observations, the sides closer to the anchor nodes exhibited poorer n value simulation, which may stem from algorithmic limitations. The weights learned by the BFGS algorithm are based on the RSSI relationships between anchor nodes to infer the target node's n value. The distances between these anchor nodes are typically larger, and their signal propagation characteristics may represent the macroscopic or average attenuation patterns of the field. However, within a very small range around a specific

anchor node, signal propagation might involve highly localized, non-linear attenuation behavior, deviating from the average pattern of the area. This discrepancy leads to inaccurate inference of the n value for nearby target nodes.

Table 4.2 Performance Analysis of n Value and Distance Estimation for Selected Test Points. (a) High Accuracy Points, (b) Lower Accuracy Points.

(a) High Accuracy Points

Eval	Loc	Item	Real n /Avg sim n /Acc(%)			Real $d(m)$ /Avg $d(m)$ /Acc(%)			Total MAE(m)
Top4	T2	d1	1.87	1.89	98.02	5.63	5.52	97.0	0.58
		d2	1.32	1.48	85.64	5.85	4.98	80.75	
		d3	3.2	3.13	97.38	14.11	14.96	92.66	
		d4	2.24	2.43	91.58	14.39	11.74	81.65	
	T6	d1	3.16	2.88	91.2	10.73	13.51	73.99	1.05
		d2	1.75	1.91	91.12	11.41	9.38	82.3	
		d3	3.12	2.64	84.93	7.43	7.66	96.9	
		d4	1.34	1.71	72.51	7.63	6.62	86.79	
	T8	d1	2.68	2.92	91.04	14.25	11.4	80.27	1.19
		d2	1.94	2.22	85.57	15.22	10.76	70.75	
		d3	1.61	1.43	88.82	5.91	5.27	89.33	
		d4	1.79	1.96	90.51	5.3	4.57	86.35	
	T19	d1	2.36	2.18	92.39	9.7	11.77	79	1.23
		d2	1.80	1.84	97.54	13.83	13.12	94.11	
		d3	1.78	1.92	91.92	10.3	8.69	84.45	
		d4	2.02	1.89	93.47	5.28	5.96	87.08	

(b) Lower Accuracy Points.

Eval	Loc	Item	Real n/Avg sim n/Acc(%)			Real d(m)/Avg d(m)/Acc(%)			Total MAE(m)
Last4	T5	d1	3.19	2.78	87.08	9.07	12.63	60.72	3.39
		d2	2.37	2.15	90.86	9.7	12.19	74.25	
		d3	3.99	3.23	81.13	8.9	14.85	33.10	
		d4	1.40	1.81	59.78	9.14	4.84	53.03	
	T10	d1	2.25	2.11	93.58	9.26	8.24	88.98	4.23
		d2	3.68	2.06	55.99	3.35	8.67	0	
		d3	2.33	2.45	95.16	12.41	11.07	89.26	
		d4	1.93	2.49	70.59	15.48	8.36	54.03	
	T16	d1	4.19	1.92	45.94	2.66	8.4	0	3.98
		d2	1.57	1.63	96.05	9.5	8.79	91.82	
		d3	1.58	1.96	75.9	15.32	9.05	59.1	
		d4	1.34	1.82	63.85	12.69	6.53	51.45	
	T18	d1	2.57	2.11	82.11	7.25	11.2	45.46	3.66
		d2	1.37	1.58	84.62	12.05	8.65	71.79	
		d3	1.15	1.63	58.34	11.66	5.67	48.63	
		d4	2.56	2.26	88.37	7.71	10.09	69.08	

Overall, the adaptive path loss model proposed in this study demonstrated higher positioning accuracy across most test points. The comprehensive test results for 20 target locations showed an average absolute error (MAE) of 2.33 meters, with a single localization operation requiring 30 milliseconds. Although larger errors persisted in some points affected by specific environmental obstacles or situated in extremely close-

range micro-environments, the method generally enhanced positioning accuracy and proved efficient in terms of time cost. This validates the feasibility of optimizing RSSI-based localization through adaptive n value inference.

4.3 Method Comparison

To comprehensively evaluate the performance of the indoor positioning system proposed in this study, this section compares it with two common and representative indoor positioning methods: first, a model-based calibration method, and second, a scene analysis-based fingerprinting method.

4.3.1 Overview of Compared Methods

To compare with other positioning methods, we randomly collected 10 new points in the experimental area to serve as independent target nodes for evaluating these positioning methods, ensuring fairness and objectivity in the comparison. The coordinates of these test points are shown in Figure 4.3, with the red dots indicating the locations of the 10 new target nodes.

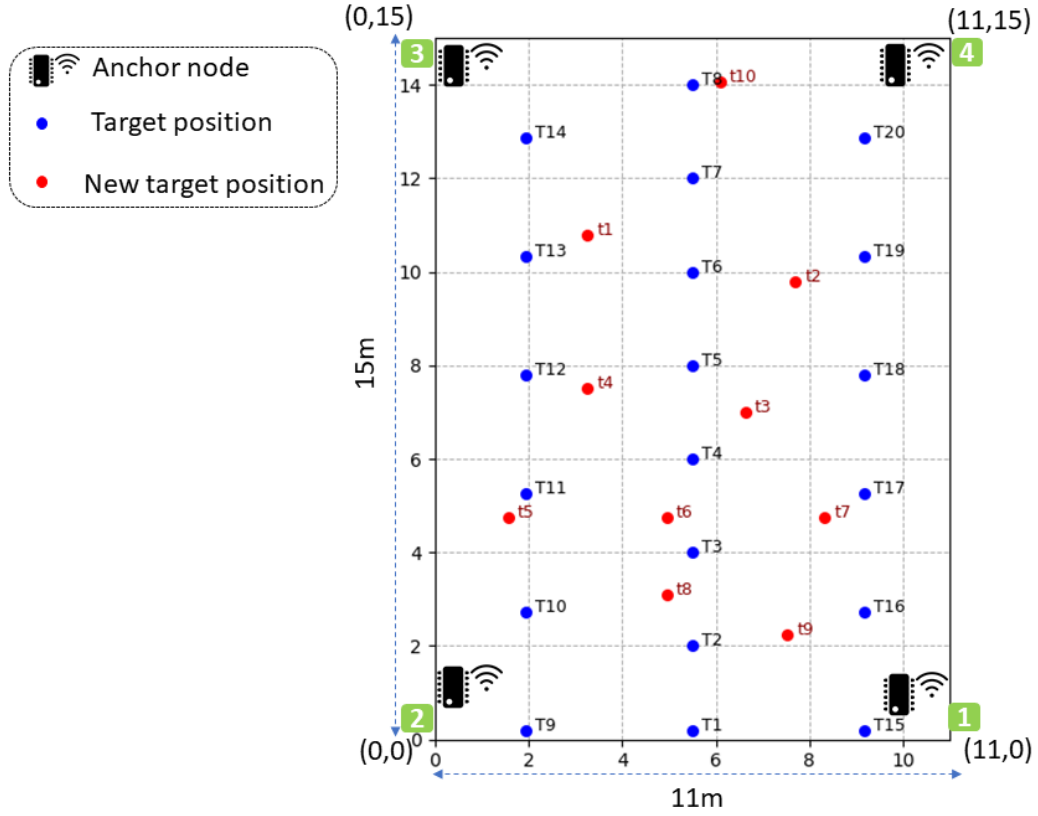


Figure 4.3 Target Location Distribution

The first method compared in this study is the Calibration Method, a commonly used approach in traditional model-based positioning [42]. Its core principle is to determine the parameters of the path loss model through extensive on-site data collection. In this study, the calibration procedure involves collecting RSSI signals at distances of 1m, 3m, 5m, 7m, and 9m from each of the four anchor nodes. Using these known distances and corresponding RSSI values, the path loss model parameters—namely RSSI at reference distance d_0 and the path loss exponent n —are estimated through nonlinear least squares fitting to construct the path loss model at each anchor location. Table 4.3 shows the fitted path loss models at the four anchor nodes. Figure 4.4 presents the fitted curve for the second anchor node.

Table 4.3 Fitted Path Loss Models at the Four Anchor Nodes

AN	Item	1m	3m	5m	7m	9m
1	<i>RSSI (dBm)</i>	-28.7	-38.1	-42.45	-45.56	-47.91
	<i>n</i>	2.1	2.5	2.8	3.05	3.25
$RSSI = -28.70 - (10 * 2.94 * \log_{10}d)$						
2	<i>RSSI (dBm)</i>	-19.62	-33.03	-38.87	-42.58	-45.66
	<i>n</i>	1.3	1.96	2.39	2.62	2.75
$RSSI = -19.62 - (10 * 2.81 * \log_{10}d)$						
3	<i>RSSI (dBm)</i>	-41.46	-50.9	-55.12	-57.72	-59.59
	<i>n</i>	2.1	2.6	2.9	3.1	3.3
$RSSI = -41.46 - (10 * 2.94 * \log_{10}d)$						
4	<i>RSSI (dBm)</i>	-28.33	-37.02	-41.6	-44.77	-47.09
	<i>n</i>	1.3	1.5	1.65	1.75	1.82
$RSSI = -28.33 - (10 * 1.71 * \log_{10}d)$						

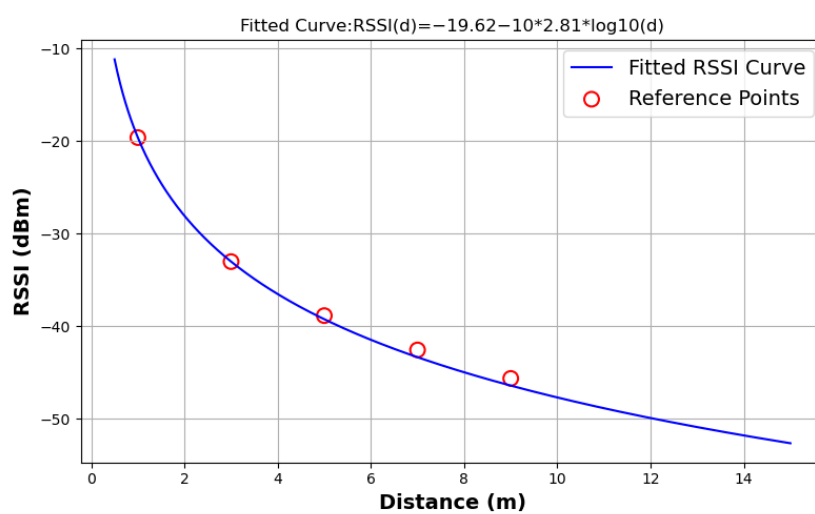


Figure 4.4 Fitted Curve of the Second Anchor Node

The second method is Fingerprinting, a typical scene analysis-based approach, which is divided into two stages: an offline stage (database construction) and an online stage (position matching), as illustrated in Figure 4.5. In the offline stage, this study uses previously collected data from 20 locations to train the fingerprinting model. The training data consists of 400 RSSI samples collected from each of the four anchor nodes, across 20 locations. The data is processed using Z-score based outlier removal, followed by feature extraction using a sliding window of size 30 and step size 10. Statistical features including mean, standard deviation, maximum, minimum, and median are extracted to serve as input for machine learning models, thereby expanding the training dataset and enhancing model accuracy. Several machine learning models are evaluated, including K-Nearest Neighbors (KNN), Random Forest (RF), XGBoost (XGB), and Support Vector Regression (SVR). In the online stage, the performance of the fingerprinting model is evaluated using data from 10 new target locations. The test data consists of 150 RSSI samples collected from each of the four anchor nodes, across 10 locations. The preprocessing steps are consistent with those in the offline stage, and the processed fingerprint data is then fed into the trained model to infer the target position.

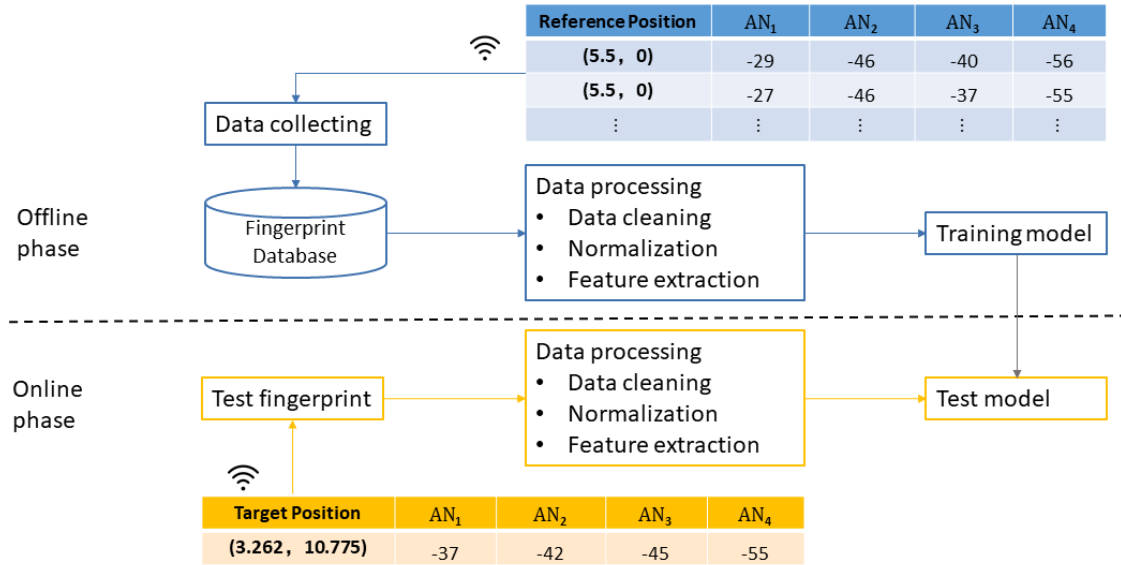


Figure 4.5 Flowchart of fingerprint method.

4.3.2 Comparison with Calibration Method

The proposed adaptive method in this study was compared with the traditional calibration method at the aforementioned 10 new test points. The evaluation results are summarized in Table 4.4. The proposed method achieved a mean absolute error (MAE) of 1.88 meters. In contrast, the calibration method resulted in a higher MAE of 3.25 meters. This indicates that the proposed method outperforms the calibration method by approximately 42% in terms of average positioning error. As shown in Figure 4.6, the proposed method yielded lower positioning errors at 9 out of 10 test locations, with the exception of the 5th point, where the error was slightly larger.

Table 4.4. Performance Comparison: Proposed method vs. Calibration Model

	Min Error(m)	Max Error(m)	Overall MAE(m)	CDF 50%	CDF 90%
proposed method	0.61	4.45	1.88	1.77	3.36
calibration model [42]	1.02	5.31	3.25	3.23	5.02

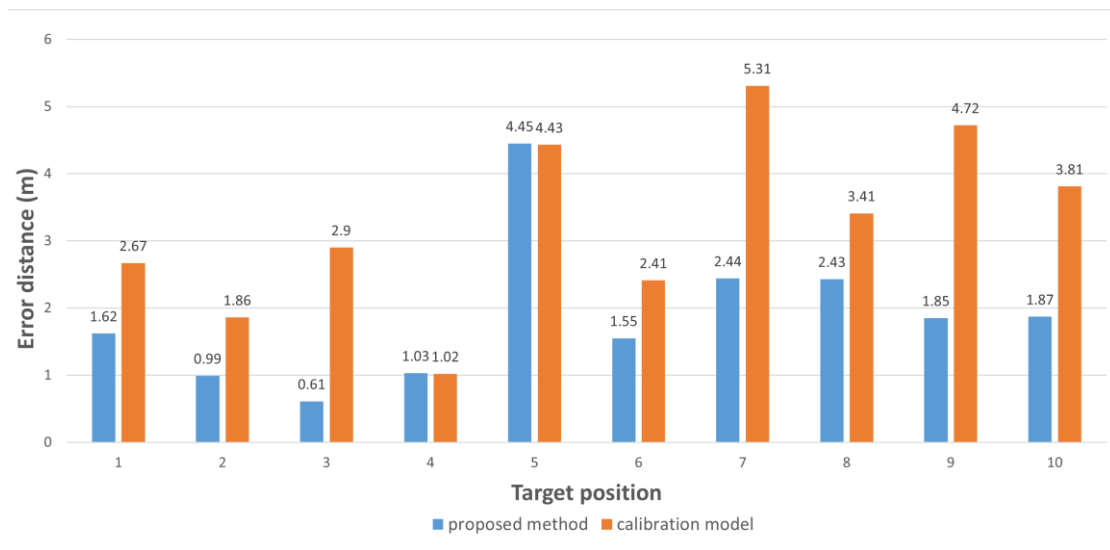


Figure 4.6 Positioning Error Comparison: Proposed Method vs. Calibration Model by Target Position

From the cumulative distribution function (CDF) plot shown in Figure 4.7, the curve of the proposed method lies above and to the left of that of the calibration method. This indicates that at any given error threshold, the proposed method achieves a higher percentage of positioning accuracy. In terms of median error (P50), 50% of the positioning errors using the proposed method are within 1.77 meters, while the calibration method requires an error of 3.23 meters to reach the same percentage. Both the median error and the 90% error demonstrate the superior performance of the proposed model.

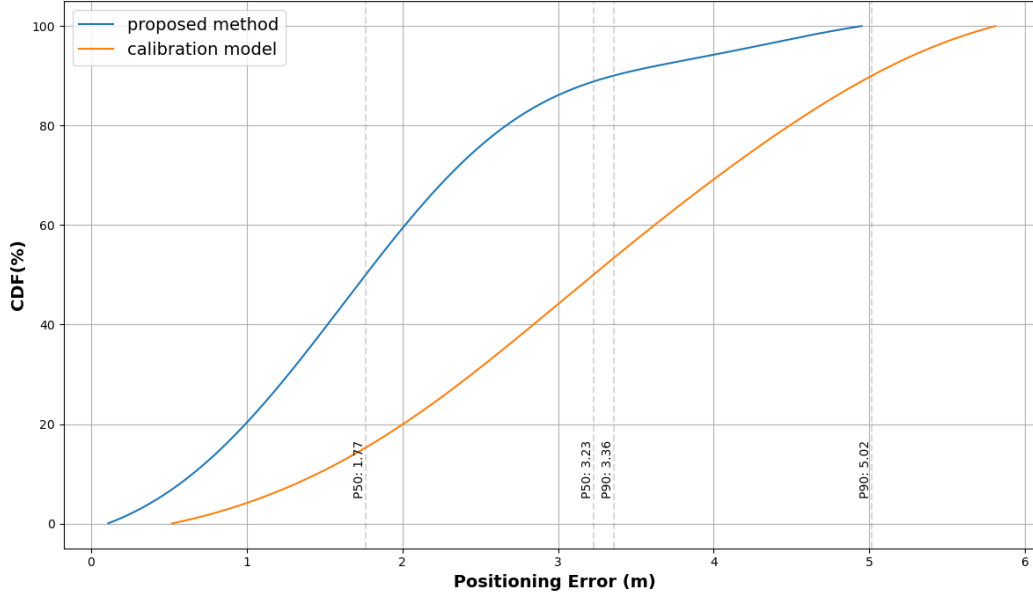


Figure 4.7 CDF of Proposed Method and Calibration Model

These results demonstrate that the adaptive path loss model proposed in this study can achieve higher positioning accuracy than the traditional calibration method, without requiring extensive on-site calibration in advance. This significantly reduces the complexity of system deployment and maintenance costs.

4.3.3 Comparison with Fingerprinting Method

In this section, the fingerprinting method is evaluated at 10 new test points to compare the positioning performance of different machine learning models, including K-Nearest Neighbors (KNN), Random Forest (RF), XGBoost (XGB), and Support Vector Regression (SVR). These algorithms have been widely applied in RSSI-based indoor positioning studies [43][44][45]. As shown in Table 4.5, among all models, the Random Forest (RF) model achieved the best overall positioning performance in the fingerprinting application, with the lowest mean absolute error (MAE) of 2.65 meters. The specific positioning errors of different machine learning models at the 10 test points are

illustrated in Figure 4.8. The CDF curves are shown in Figure 4.9, where the Random Forest (RF) model curve lies furthest to the upper left, indicating that it covers a higher proportion of positioning points within the same error range compared to the other models.

Table 4.5 ML-Based Fingerprinting Model Performance Comparison

	Min Error(m)	Max Error(m)	Overall MAE(m)	CDF 50%	CDF 90%
KNN(n=6)	1.19	5.2	2.98	2.81	4.74
RF	1.27	4.63	2.65	2.55	4.07
XGB	1.15	4.66	2.74	2.65	4.45
SVR	1.77	5.61	3.18	3.14	4.97

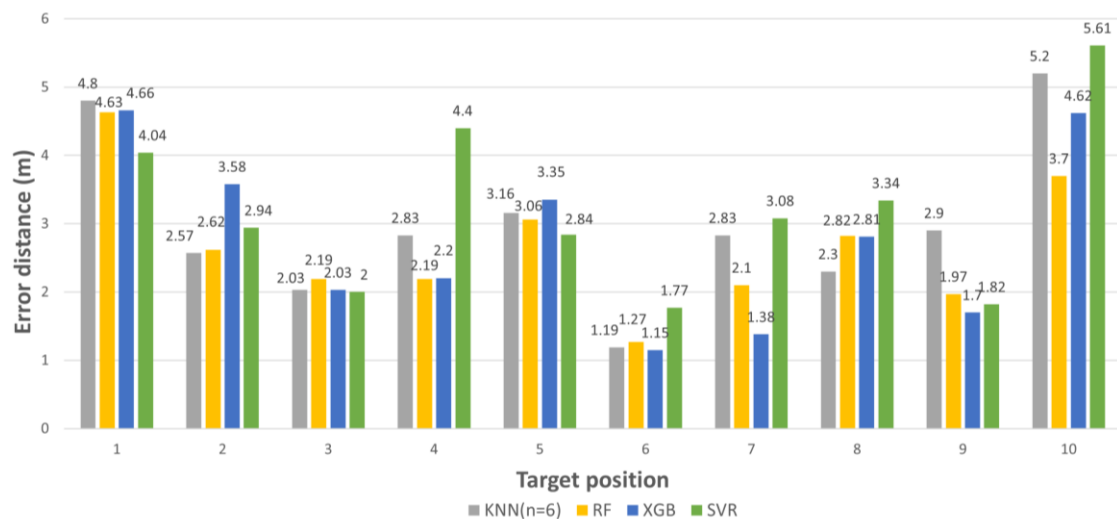


Figure 4.8 Positioning Error of ML Models in Fingerprinting by Target Position

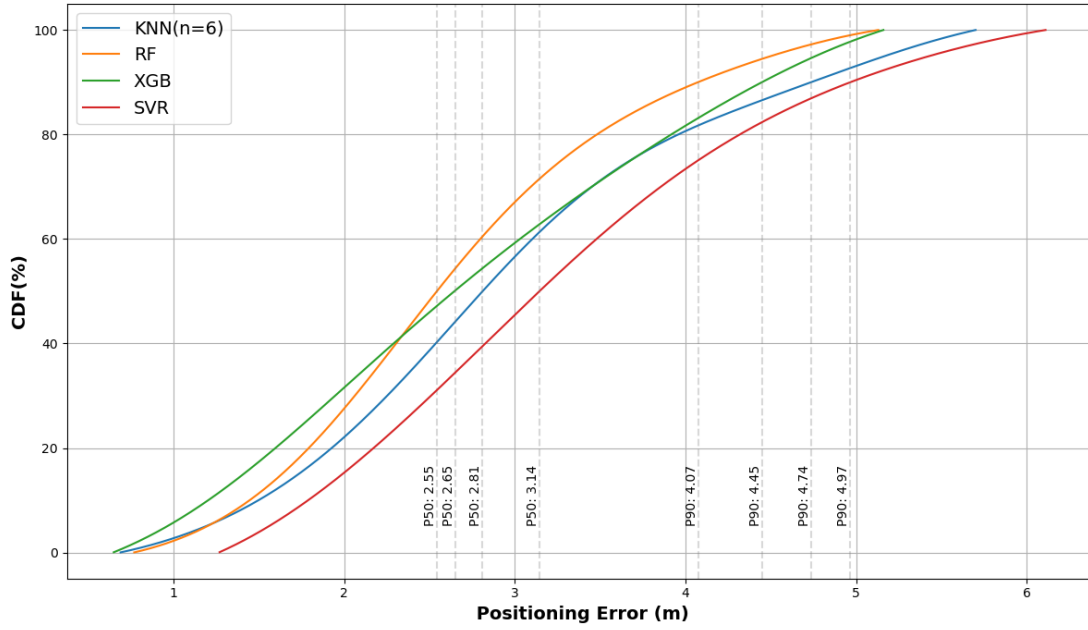


Figure 4.9 CDF of ML-Based Fingerprinting Model

Although the RF model performed best among the fingerprinting methods, the overall accuracy of fingerprinting was still significantly inferior to the adaptive method proposed in this study. This is mainly due to the sparsity of the training data used in the experiment. In an area of 11m x 15m, only 20 reference points were used for training, which may be insufficient to fully capture the signal distribution characteristics of the entire environment. As a result, when the model attempts to estimate positions in areas not effectively covered by reference points, the positioning accuracy degrades. This experimental outcome also highlights the inherent limitations of fingerprinting methods, in which positioning accuracy is highly dependent on the density of reference points. To achieve usable or high-precision results, extensive effort is required in the early stages to construct a dense fingerprint database. Moreover, as the environment changes, the fingerprint database must be frequently maintained and updated, significantly increasing both the deployment and long-term operational costs and complexity.

4.3.4 Comprehensive Performance Comparison

This section presents a comprehensive performance comparison of various indoor positioning methods, including two model-based approaches and four machine learning models used in fingerprinting.

Table 4.6 summarizes the overall performance of different positioning methods. The method proposed in this study achieved a mean absolute error (MAE) of only 1.88 meters, which is significantly lower than that of all other models: 3.25 meters for the calibration method and 2.65 meters for the best-performing fingerprinting model, Random Forest (RF). This clearly demonstrates the substantial advantage of the proposed method in terms of average positioning accuracy. This advantage is also evident in the cumulative distribution function (CDF) curves shown in Figure 4.10.

Table 4.6 Performance Comparison of All Methods

	Min Error(m)	Max Error(m)	Overall MAE(m)	CDF 50%	CDF 90%
proposed method	0.61	4.45	1.88	1.77	3.36
calibration model	1.02	5.31	3.25	3.23	5.02
KNN(n=6)	1.19	5.2	2.98	2.81	4.74
RF	1.27	4.63	2.65	2.55	4.07
XGB	1.15	4.66	2.74	2.65	4.45
SVR	1.77	5.61	3.18	3.14	4.97

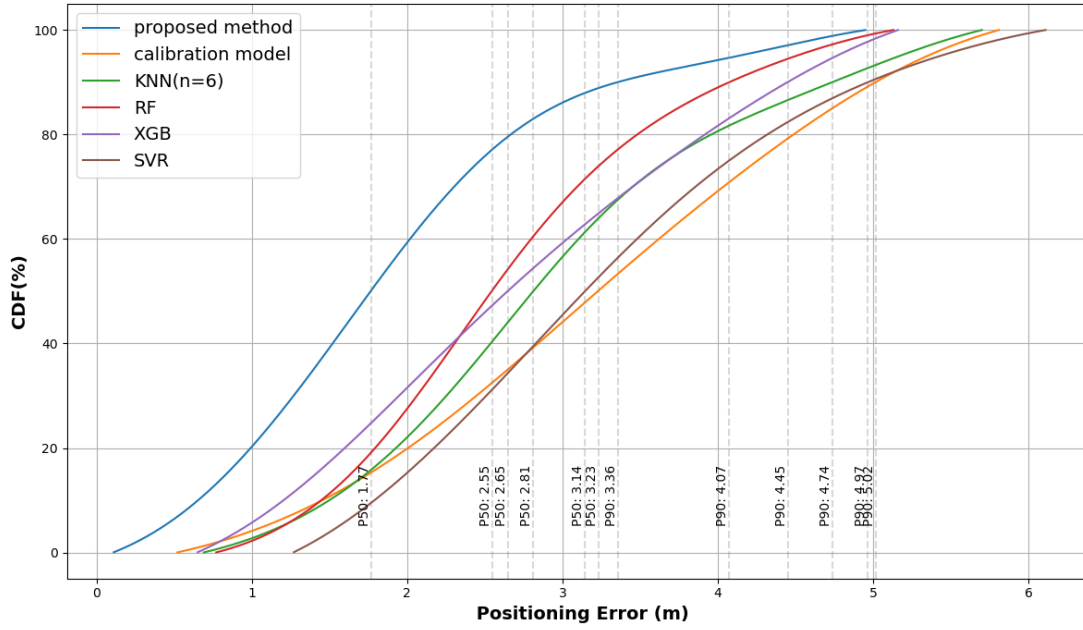


Figure 4.10 CDF of All Methods

In addition to numerical accuracy, the method proposed in this study also demonstrates significant advantages in terms of practicality. Compared to the calibration method, it does not require large-scale and time-consuming on-site data collection and parameter fitting. Meanwhile, unlike the fingerprinting method, it avoids the need to construct a large fingerprint database, minimizes sensitivity to environmental changes, and eliminates the high maintenance costs. These attributes make the proposed method more attractive in terms of deployment flexibility and cost-effectiveness.

4.4 Comparison with Related Work

To further validate the performance and academic contribution of the proposed method, this section compares it with recent related studies based on similar principles. In particular, we focus on a paper published in 2024 in *IEEE Access* by Dariusz Janczak et al.[21], which proposed an indoor positioning method based on Bluetooth Low Energy (BLE) and Received Signal Strength Indicator (RSSI). The comparison results are shown in Table 4.7.

Both studies employed RSSI-based multilateration techniques, with RSSI data collected through real-world experiments. Regarding the experimental setup, the referenced study used 5 anchor nodes in an environment of approximately $11\text{m} \times 23\text{m}$ and tested 13 target positions. In contrast, this study used 4 anchor nodes in a smaller area of approximately $11\text{m} \times 15\text{m}$ and tested 30 target positions. In terms of accuracy enhancement, the referenced paper used a simple moving average RSSI estimator combined with an on-site calibrated distance model to improve distance estimation accuracy. Meanwhile, this study applied Kalman filtering for RSSI signal smoothing and improved distance estimation through a simulated path loss exponent n . Both studies used Mean Absolute Error (MAE) as the primary error evaluation metric. The referenced paper reported a final average positioning error of 2.5 meters, while the proposed method in this study achieved a lower average error of 2.11 meters across 30 test positions.

Table 4.7 Comparison with The Results of Different Studies

	Indoor Positioning Based on Bluetooth RSSI Mean Estimator for the Evacuation Supervision System Application <i>2024, IEEE Access</i>	Proposed method
Dataset	Collected by experiments (5 anchor nodes)	Collected by experiments (4 anchor nodes)
Area size	11m*23m	11m*15m
Methods	RSSI-based Multilateration RSSI mean estimator calibration mode	RSSI-based Multilateration Kalman Filter Simulate n method
Test position	13	30
Result	MAE:2.5m	MAE:2.11m

Through the comparison with this recent 2024 study, it is evident that the indoor positioning method based on the adaptive path loss model proposed in this research achieves superior positioning accuracy despite using fewer anchor nodes and testing more target locations. This demonstrates the potential of the proposed method to effectively improve indoor positioning accuracy while reducing reliance on extensive calibration efforts and the number of wireless communication devices. It represents a valuable contribution to the advancement of research in the field.

Chapter 5 Conclusions and Future Works

5.1 Conclusions

This study addresses the issues of low positioning accuracy and high dependence on prior calibration in RSSI-based indoor positioning methods by proposing an adaptive path loss model. The system automatically fits the path loss exponent n at each point, reducing the need for manual calibration while improving positioning accuracy. Utilizing Wi-Fi as the sensing medium, four ESP32 modules were deployed as anchor nodes in an $11\text{ m} \times 15\text{ m}$ indoor environment to collect large volumes of RSSI data, with Kalman filtering applied to remove anomalies and noise.

The core workflow consists of four steps: (1) calculating the actual path loss exponent n between anchor nodes, (2) using the RSSI values between anchor nodes to fit an adaptive n value for each target point, (3) estimating the distance to each anchor node, and (4) applying the least squares method to determine the target coordinates. Experimental results show that the proposed method achieved an average positioning error of 2.33 meters when tested on 20 target points. Further testing on 10 independent target points yielded an even lower average error of 1.88 meters, significantly outperforming both the calibration method and the fingerprinting approach, while also avoiding time-consuming data collection efforts. The CDF curve clearly demonstrates the superior overall performance of the proposed system. Compared with a recent study [21], the proposed method achieved a better average error of 2.11 meters despite using fewer anchor nodes and testing more locations. Overall, the system offers an efficient

and accurate indoor positioning solution while reducing deployment and maintenance costs.

5.2 Future Works

The proposed adaptive path loss model has been validated in a single indoor test environment, proving its effectiveness in reducing positioning errors and calibration needs. However, to further enhance the system's real-time capability, generalizability, and application potential, future improvements may focus on expanding the deployment area, upgrading hardware, refining the user interface, and optimizing the model. These enhancements aim to improve the system's usability and performance across various application scenarios, achieving a scalable, cost-effective, and high-precision indoor positioning solution.



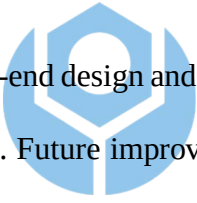
■ Expansion of Experimental Environments to Validate Model Generalization

This study was conducted in the lobby of the Sheng-Yan Building at Fu Jen Catholic University (11 m × 15 m), an environment featuring obstacles such as columns and iron doors, with data collected under various times and conditions. However, the adaptability of the model in more complex or structurally diverse environments remains to be validated. Future work may involve conducting experiments in different building types and materials, such as office buildings, hospitals, warehouses, and shopping malls. Further extensions to multi-floor and highly dynamic environments could help evaluate the system's stability and accuracy under varying RSSI propagation conditions, thereby enhancing its generalizability.

■ Adoption of Low-Power and Fast-Response Communication Modules

The current system uses ESP32 Wi-Fi modules for RSSI collection, which, while stable, have relatively long scan times and higher power consumption. Future implementations could consider incorporating Bluetooth Low Energy (BLE) modules, especially Nordic's nRF series chips. BLE offers advantages such as fast scanning, low power consumption, high deployment density, and support for Angle of Arrival/Departure (AoA/AoD), making it possible to reduce positioning latency and improve accuracy. Although BLE has a shorter transmission range and may require more anchor nodes, its low cost and energy efficiency make it a worthwhile investment. Future work will assess the feasibility and performance trade-offs of integrating BLE modules into the system.

■ **Development of Mobile Interface to Enhance Usability**



This study focused on the back-end design and validation of the positioning model, without integrating a user interface. Future improvements may include developing an intuitive mobile application to support real-time location display, indoor navigation, location sharing, data visualization, and user feedback mechanisms. Such features could significantly enhance system usability and support integration with the Internet of Things (IoT) and Location-Based Services (LBS), thereby boosting its practical applicability and commercialization potential.

■ **Improving Model Performance at Short Distances**

Experiments revealed that the model's accuracy in predicting the path loss exponent n is lower at locations very close to anchor nodes, resulting in increased positioning errors. This issue may stem from the limited applicability of the log-distance model

under near-field conditions. Future improvements could include hybridizing the free-space and log-distance models, with mode switching based on distance, or introducing distance-based weighting into the BFGS optimization to ensure the estimated n better reflects local environmental characteristics. Additionally, modeling n as a function or enhancing the density of near-distance training samples could help improve learning accuracy. These strategies are expected to enhance the model's robustness and accuracy in short-range positioning tasks.



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